Urban Transit Infrastructure and Inequality*

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March 2023

Abstract

How does transit expansion impact welfare and inequality in a city? Using the universe of transit farecard trips from Singapore, we find that low-income workers are more sensitive to changes in travel time than their high-income counterparts. Since the former are overwhelmingly employed in non-tradable sectors, changes in consumption travel could induce a spatial re-organization of low-income jobs in the city, with important distributional implications. This re-organization is investigated through an urban spatial model with heterogeneous workers, incorporating travel to work and to consume non-tradable goods and services. We find that the Downtown Line resulted in large welfare gains for high-income workers, but near zero for low-income workers. All workers benefited from improved access to consumption opportunities, but low-income jobs in the non-tradable sector moved to less attractive workplaces. Abstracting from consumption travel underestimates the inequality effects five-fold.

*We thank Pol Antras, Costas Arkolakis, Nick Buchholz, Antonio Ciccone, Edward Glaeser, Nathan Hendren, Kate Ho, Adam Kapor, Matthew Kahn, Jakub Kastl, Gabriel Kreindler, Jeffrey Lin, Marc Melitz, David Nagy, Christopher Neilson, Stephen Redding, Nick Tsivanidis, Matthew Turner, participants of the NBER Summer Institute: Urban Economics 2021, 15th North American Meeting of the Urban Economics Association, and the 10th European Meeting of the Urban Economics Association, and a number of other colleagues, conference and seminar participants for helpful comments and suggestions. We thank the Singapore Land Transport Authority for providing data to the corresponding author. Finally, we thank the Princeton Institute for International and Regional Studies, as well as the Department of Economics at Princeton University, for financial support. Responsibility for results, opinions and errors is the authors' alone.

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1 Introduction

With 68% of the world population projected to live in urban areas by 2050, governments will continue to spend vast sums on expanding mass transit (United Nations 2018). Yet, as cities worldwide grow more and more unequal, a question beckons: How are the economic gains from transit expansion shared between low- and high-income workers?[1] Existing research focuses on differential access to employment opportunities across worker groups emphasizing the cost of travel to work, particularly in the developing world (Tsivanidis 2019; Balboni et al. 2020). However, a large share of urban trips are unrelated to work. These trips are taken for the consumption of non-tradable goods and services, including visiting restaurants, retail stores, and salons.[2]

In this paper, we leverage the universe of transit farecard trips from Singapore around the opening of the Downtown Line (DTL, described in Section 3). These farecard data are combined with administrative spatial data on land use, rents, expenditures and employment. The data speak to a novel quantitative spatial model, incorporating heterogeneous worker types and travel to consume non-tradable goods. We seek to shed light on the interplay between four pieces of the puzzle: transit expansion, consumption travel, inequality, and the spatial re-organization of economic activity. The core message of our paper is that accounting for travel related to the consumption of non-tradables is critical for evaluating the impact of public transit expansion on inequality, especially given that low- and high-income workers live, work, and consume in different locations. Since low-income workers are overwhelmingly employed in non-tradable sectors (as waiters, retail workers, manicurists and so on), changes in consumption travel could induce a spatial re-organization of low-income jobs in the city, with important implications for inequality.

In Section 4, we first provide suggestive evidence that low- and high-income workers live, work, consume, and travel differently. In Singapore, a closed city, low-income workers earn a monthly salary below the 25th percentile, and are explicitly targeted for transit subsidies and income support. Next, we consider the impact of the DTL, the focus of the remainder of our empirical analysis. Constructed at a cost of US$15.5 billion, the DTL is the longest underground mass rapid transit line in Singapore, and is the city-state's

[1] In particular, there is concern that low income workers face much larger commuting costs than high income workers. See: "Long Commutes Are Awful, Especially for the Poor", The Atlantic; "Public Transit Cuts Felt Deepest in Low-Income Areas", NYTimes; "How long commutes worsen inequality", CBS News.

[2] According to the National Household Travel Survey, as of 2017, in the United States, 31% of travel miles are made for work purposes, while 33% of travel miles are made for shopping or meals and another 26% for social or recreational purposes. See Appendix Figure A2 (Department of Transportation 2017). Additionally, in 2018, in Singapore, non-tradable consumption makes up over 50% of total household expenditure (Department of Statistics 2019). See Appendix Figure A3.

most ambitious public transit project to date (The Straits Times 2017). The DTL sought to provide the north-west and eastern areas a direct link to the center of the city. We ascertain that the 2008 announcement of the precise locations of DTL stations coincided with higher appreciation for apartments near each DTL station, relative to those farther away. Additionally, when Stage 2 of the DTL opened in December 2015, we found that low- and high-income workers adjusted their travel patterns differently in both the short run and up to a year after the line opened.

Motivated by the descriptive evidence, in Subsections 6.1 and 6.2, we utilize gravity regressions to assess how responsive each type of worker is to increases in travel time. We find that low-income workers have larger travel elasticities than high-income workers, and elasticities are larger than for consumption than for work travel across both types. These findings are robust to a differences-in-differences analysis comparing origin-destination pairs treated by the line opening (Severen 2019), and accounting for sparsity in the matrix of said origin-destination pairs (Dingel and Tintelnot 2020). In the pre-DTL equilibrium, low-income workers commute shorter distances than high-income workers do, but travel as far as the latter to consume non-tradables. While our gravity regressions give us a framework to understand the data, they leave us slightly short: to assess how the DTL impacts welfare and inequality in Singapore, we need more structure to understand what happens to the spatial distribution of economic activity in the city.

To interpret these empirical findings, we develop a quantitative spatial model, described in Section 5, that nests several others. While our model incorporates heterogeneous worker types by income (low/high), our main theoretical contribution is modeling travel to consume non-tradable goods in a manner that is informed by the data and admits tractable counterfactuals using exact-hat algebra (Dekle, Eaton, and Kortum 2008). In the model, workers travel to work and to consume non-tradable goods and services. The city in which they reside is modeled as a set of neighborhoods, with a transportation network characterized by bilateral travel times between locations. Worker groups differ in their preferences over residences and consumption locations, productivities over workplaces and sectors, consumption patterns, and travel costs and elasticities. Workers first choose where to live, then where to work and where to consume non-tradables.

When choosing where to live, each worker trades off rents against residential amenities, as well as access to employment and consumption opportunities. When choosing where to work, each worker evaluates their type-specific wages, match-specific productivities and commute distances. Finally, when choosing where to consume non-tradables, they consider prices, type-specific idiosyncratic consumption amenities, and travel distances. On the production side, the non-tradable and tradable sectors have different in-

put requirements over commercial floor space and labor provided by each worker group, with the non-traded sector being much more intensive in low-income labor. Market clearing in the non-traded sector implies that, in each location, consumption travel patterns drive the demand for non-traded production, and hence also the demand for low-income workers. Finally, the model also accommodates agglomeration forces, and allows for type-specific spillovers in endogenous residential amenities. These spillovers thus drive segregation in the city.

With the model in hand, to undertake our counterfactuals, we estimate and calibrate several other key parameters in Section 6. Through the lens of the model, the fixed effects in our gravity regressions can be interpreted as measures of location attractiveness. These measures are validated against external data on associated amenities. Model-implied average wages by worker type and residence are also highly correlated with those observed in administrative data. Spillovers in residential amenities are larger for low-income workers, and residential elasticities are slightly larger for high-income workers. Finally, using changes in travel times as observed in the farecard data before and after the opening of the line, we find that the model predicts well post-DTL travel flows, residential patterns and the entry of firms in the non-tradable sector.

After having computed our estimates, using exact-hat algebra, we quantify the welfare and inequality effects of the DTL in Section 7. We find that the DTL improves welfare for high income workers by 3.15%. However, low income workers experience near zero net benefits. Although both groups can better access consumption opportunities, low-income workers see their jobs (in non-tradable production) moving to places they deem less attractive, offsetting their gains from the consumption channel. This disparity between the rich and the poor is driven by two mechanisms. First, the DTL disproportionately improves access for the many high-income areas served directly by the line. Second, more consumers consume non-tradables near the city center because the DTL has made it easier for them to go downtown. Thus, non-tradable low-income jobs move to the center of the city, yet low income workers still live away from the center. Average commute times increased 1.5% for low income workers, while high income workers saw a 1% reduction.

Abstracting away from travel to consume non-tradables, we find a five-fold underestimation of the inequality effects of the DTL. Absent non-tradable consumption, we see that both worker groups benefit but a slightly larger share of the gains go to high-income workers. Thus it is quantitatively important to capture the spatial re-organization of low-income non-tradable sector jobs. Aggregate welfare gains are also underestimated by about 40%, stemming from ignoring improved consumption access.

Additionally, we consider the role of line placement in driving our results. We sim-

ulate the removal of the North-South Line (NSL), a trunk line primarily serving more diverse neighborhoods. When the NSL is removed, welfare falls for both low and high types, by 4.41% and 2.08% respectively. Both types experience reductions in consumption and employment access, and inequality rises by 2.33%. Thus, line placement matters when considering the implications of transit expansion on welfare and equity.

Finally, we conduct a model decomposition exercise. We find that low-income workers spend less on non-tradables; thus they gain less from better consumption opportunities. They also have lower travel costs and higher travel elasticities; so they can better substitute to more attractive work and consumption locations, therefore benefiting less from shorter trips. Yet these low-income workers have relatively more dispersed residential preferences, so they are less able to move their residences to capitalize on improved access elsewhere.[3]

2 Literature Review

This paper speaks directly to the literature studying the impact of transportation infrastructure within cities (McDonald et al. 1995; Gibbons and Machin 2005; Baum-Snow and Kahn 2005; Glaeser, Kahn, and Rappaport 2008; Billings 2011; Donaldson and Hornbeck 2016; Severen 2019; Donaldson 2018; Brooks and Lutz 2019; Gupta, Van Nieuwerburgh, and Kontokosta 2020; Heblich, S. Redding, and D. Sturm 2020). The most closely related papers to ours are Tsivanidis (2019) and Balboni et al. (2020), who study the Bus Rapid Transit (BRT) systems focusing on inequality in Colombia and Tanzania respectively. Both papers estimate quantitative spatial models restricted to workplace commuting, finding small inequality effects. We innovate on the literature by incorporating the role of travel to access non-tradable goods and services. These travel costs are crucial to evaluate the distributional welfare effects of transit expansions. Consistent with prior work, when we abstract from consumption trips, we find the DTL barely affects inequality. However, in our setting, factoring in consumption trips gives us the opposite conclusion.

Our paper also contributes to several other strands of literature. The first is the growing body of work on quantitative spatial models more generally (S. Redding and Daniel M. Sturm 2008; Allen and Arkolakis 2014; Ahlfeldt et al. 2015; S. Redding 2016; Allen, Arkolakis, and Li 2016; Caliendo et al. 2018; Desmet, Nagy, and Rossi-Hansberg 2018;

[3]In Online Appendix F, we also evaluate the cost effectiveness of the DTL via the Marginal Value of Public Funds measure (Hendren and Sprung-Keyser 2020). Combining construction and operating cost data with estimates of willingness to pay from our quantitative model, we find that the DTL provides 8.65 dollars of benefits per dollar of government spending. The social benefit-cost ratio is 5.28. Although these benefits largely accrue to high-income workers, the DTL remains highly cost effective.

Monte, S. J. Redding, and Rossi-Hansberg 2018; Adao, Arkolakis, and Esposito 2019; Allen and Arkolakis 2022). Our paper is closely related to that of Miyauchi, Nakajima, and S. Redding (2022), who explore the role of trip chaining in explaining consumption externalities across locations in Tokyo.[4] Unlike theirs, we are the first to explore inequality in work and consumption access via a quantitative spatial model. This paper also speaks to quantifying the value of urban amenities (Roback 1982; Blomquist et al. 1988; Glaeser, Kolko, et al. 2001; Albouy et al. 2016; Cragg and Kahn 1997; Bayer et al. 2009, Fan, Guthrie, and Levinson 2016; Diamond 2016; S. J. Redding and Daniel M Sturm 2016; Almagro and Domínguez-Iino 2020). We demonstrate the important role of non-tradable retail amenities in shaping inequality and the spatial distribution of economic activity in the city. Moreover, we are also able to quantify differences in preferences over amenities across worker groups. Finally, our paper is related to the literature on inequality in cities (Brueckner et al. 1999, Glaeser, Resseger, and Tobio 2009; Su et al. 2018; Fogli and Guerrieri 2019; Couture et al. 2019). We show that changes in travel access can have large implications for urban inequality.

3 Background and Data

3.1 Context: Singapore and the Downtown Line

Singapore is an island city-state in Southeast Asia. With a population of about 6 million inhabitants, Singapore is the third densest country in the world, and is among the densest cities (World Bank 2019).[5][6] Inequality in the city-state is high; with a Gini coefficient of 46.4, Singapore ranks only behind Hong Kong and the United States among high-income countries.

The population in Singapore is heavily reliant on public transportation, comprising buses, light-rail and mass-rail networks. The Singapore government heavily restricts the supply of cars with a Vehicle Quota System. As of 2018, the price of a Volkswagen Golf 1.4 is US $110,479.80, 5 times more expensive than in the United States (US$21,845).[7] As a

[4]In our setting, we abstract away from trip chaining because we do not directly observe the "stays" each commuter makes, and we face a curse of dimensionality over trip itineraries.

[5]There are highly restrictive immigration policies with strict foreign worker quotas, and limited out- and in- migration (Ministry of Manpower 2020).

[6]The spatial distribution of non-tradables across Singapore is similar to that of other major cities (see Figure A5).

[7]According to the 2019 Worldwide Cost of Living Survey carried out by the Economist Intelligence Unit, Singapore "remains the most expensive place in the world to buy and run a car." Additionally, the city's Electronic Road Pricing System imposes large tolls for driving to encourage the use of public transportation. Tolls are highest during commuting hours – a car trip from the northwest to the center of the city 8am and

result, only 10% of households in Singapore own a car, and far fewer drive to work. With the public transit system carrying over 4 million passengers per day, public transportation is the primary means of travel for the population.[8] Approximately 60% of trips are made on buses, as opposed to rail lines.

The Downtown Line is the longest underground and mass rapid transit line in Singapore. At 41.9 kilometres (26.0 miles) and with 34 operation stations, the line runs from Bukit Panjang station in the north-west to Expo station in the east via the Central Area.[9] The line was first announced on 23 October 2001, and was built in three phases. The first phase opened in December 2013, and was composed of 6 stations from Bugis to Chinatown station within the Downtown area. The second phase from Bukit Panjang to Rochor station, which linked the north-west to the center of the city, opened in December 2015. The third and final phase from Fort Canning to Expo station, connecting the east to the city of the city, opened in October 2017. The line, at a cost of 15.5 billion USD, is considered the government's most ambitious public transit project to date (The Straits Times 2017). The purpose of the line was to provide the north-west and eastern areas a direct link to the center of the city, and to alleviate congestion on various other rail and bus routes.

3.2 Data

Our primary source of data is public transit fare card data (EZ-Link) from the Land Transport Authority of Singapore.[10] We observe all trips made by public transit (mass rail, light rail or bus) linked to a individual's fare card.[11] Our data set covers one week every quarter between June 2015 and June 2018, and three full months between Dec 2015 and February 2016. The longer quarterly data set allows us to observe changes in transit patterns from before the opening of Phase 2 of the Downtown Line to after the opening of Phase 3. The full three months of data captures the period directly before and after the opening of Phase 2 of the Downtown Line. In total, we observe over a billion trips. For each trip, we observe the origin, destination, and start and end time of the trip. Each individual in our data set is categorized into eight groups: Adult, Low-Income Worker, Primary Student, Secondary Student, Tertiary Student, Senior Citizen, Military, and Person with Disabilities. In our analysis, we focus the Adult and Low-Income Worker categories. Low-Income

9am on a weekday is tolled up to 15 dollars.

[8] Appendix Table A1 provides a summary of the mass-rail lines in Singapore.

[9] See Figure A6.

[10] A heat map visualization of the data is presented in Appendix Figure A7.

[11] These fare cards are called EZ-Link cards.

Workers are those who earn a monthly salary below the 25th percentile ($2,000 SGD prior to 2020).[12]

Our unit of analysis is the *subzone*, the smallest administrative planning unit in our data sets, delineated by the Urban Redevelopment Authority of Singapore. Singapore is divided into 323 subzones with a median size of 1,229,894 square meters. These subzones are contained within larger spatial units, namely 55 planning areas and 5 planning regions. We link all bus and train stops to subzones.

We use our fare card data to generate work and consumption travel probabilities conditional on residential subzone. From a universe of 7.3 million unique fare cards, we restrict our data set to 3 million fare cards with at least 20 trips over our panel.[13] We include only Adult and Low-Income fare cards in order to capture the working population, dropping students, people with disabilities, those in the military, and the elderly. First we identify each individual's residence as the modal first origin and last destination of the day, where each person typically starts or ends the day. Next, we identify each individual's workplace as the modal destination during the morning rush hour (5am to 11 am) and origin during evening rush hour (3pm to 11pm). Finally, we classify all remaining trips as consumption trips.[14]

We also use data from several other administrative sources. The General Household Survey (GHS) 2015 from the Department of Statistics provides detailed information on population, employment, income, and demographics at the subzone level (Department of Statistics 2015). The GHS is conducted in between the Population Censuses which are conducted once in ten years covering a wider range of topics. Most data is compiled from administrative records across multiple sources, and additional information not available from administrative sources is collected from a sample survey of over 30,000 households. The Labor Force Survey 2018 provides data on employment and wages by industry (Ministry of Manpower 2018). The Household Expenditure Survey 2018 provides detailed

[12]Low Income Workers receive up to a 25% subsidy on their fare costs. The Central Provident Fund automatically determines eligibility based on tax filings and sends an individual application package informing workers of their eligibility, details of the scheme, and how to apply for the subsidized card (Ministry of Transport 2014).

[13]We obtain similar travel probabilities when we selected different trip thresholds for fare cards to be included in our data set.

[14]Despite our rich data, one concern might be that some of these trips are in fact work trips taken by individuals without a fixed employment location, or the self-employed. We argue that such individuals make up a minority of transit users. As of 2015, according to the World Bank Development Indicators Database, about 14% of labor force participants in Singapore are self-employed. In our communications with self-employed tradespeople ranging from electricians to plumbers, we found that most utilized work vans (taxed less punitively than personal vehicles) for work and leisure travel. Moreover, doctors in private practice would perform house calls in personal vehicles. Therefore, we believe that classifying the remaining trips as consumption travel is correct to first order.

data on household and worker expenditures by income bracket (Department of Statistics 2019). We classify itemized expenditures on goods and services into tradables and non-tradables. The REALIS dataset from the Urban Redevelopment Authority provides data at the subzone level on residential and commercial land use (broken down by sector) as well as average rent per square meter by commercial and residential land (Urban Redevelopment Authority 2022). Housing and Development Board (HDB) transaction data provides the universe of HDB flat sales with information on price, address, flat size, and number of rooms between 1999 and 2019 (Housing and Development Board 2019).

Finally, we also collect spatial data on amenities from the government data portal (Open Government Products 2019). First of all, we have two cross-sectional data sets, dating to 2015 and 2018, covering the universe of licensed food establishments in Singapore from the Singapore Food Authority. Second, data on all supermarkets and hawker centers in Singapore are from the National Environment Agency.[15] Third, the Ministry of Education provides data on the location of all schools. Fourth, the Singapore Land Authority provides data on all parks and community clubs. Fifth, the Singapore Ministry of Health provides data on the location of all clinics. We geocode all data and link each address or coordinates to the subzones in which they are contained.

4 Descriptive Analysis

In this section, we present reduced form results on the impact of the Downtown Line and motivating facts which guide our structural model. In all our analysis, we define low-income workers as those earning below the 25th percentile ($2,000 SGD or $1,500 USD per month, less than half of median earnings)[16] and high-income workers as those earning above, consistent with our fare card data set. Our unit of analysis is the subzone, which we will henceforth call a neighborhood.

4.1 Residential patterns

Low- and high-income groups have different residential patterns across the city. Figure 1 plots the share of high-income workers that live in each neighborhood in 2015. High-income workers primarily cluster near the center of the city (eg. Bukit Timah, Tanglin) with only some concentration in certain neighborhoods near the coasts of the island (eg. East Coast Park). On the other hand, low income workers live farther from the city cen-

[15] We present a map of the data in Appendix Figure A8.

[16] Singapore's median income in 2019 was $4,563.

ter, with many living closer to the north. Figure 1 also shows that the Downtown Line runs through some of the neighborhoods with the highest concentration of high-income workers in Singapore, clustered in or just west of the city center. The many low income residents in the north are not directly served by the Downtown Line. This suggests that high-income workers may disproportionately benefit from the DTL, since the line serves them directly.

4.2 Employment patterns

We find that neighborhoods which employ many low income workers are intensive in non-tradable sector production. Figure 2 plots the relationship between the share of commercial land used by non-tradable sectors and the share of workers that are low-income. We find a strong positive correlation as expected. Labor Force Survey data shows that 51% of workers in the non-tradables sector are low income, while making up only 25% of the labor force, as shown in Table A2.[17] Only 14% of workers in the tradables sector are low income. Appendix Figure A9 plots the share of high-income workers that work in each neighborhood. This suggests that the spatial distribution of non-tradable jobs is highly important for low-income workers.

4.3 Consumption patterns

Using itemized Household Expenditure Survey data broken down by income group, we find that high-income consumers spend more on non-tradables than low-income consumers. Figure A10 shows that high-income workers spend 67% of their income on non-tradable goods and services, 14% on tradables and the remaining on housing. In contrast, low-income workers spend just 57% of their income on non-tradable goods and services and 20% on tradables. This suggests that improving access to consumption opportunities will raise welfare more for high-income than low-income workers.

4.4 Travel patterns

We find that low income workers make shorter commutes, at a median of 25 minutes, relative to 30 minutes for high income workers. However, consumption trips are more similar in times across income groups and are shorter than workplace trips at around 23 minutes for high-income workers and 22 minutes for low-income workers respectively.

[17] Specifically, 61% of employees in the food and accommodation, 49% in retail, and 41% in personal services are low-income.

Figure 3 plots the travel time distributions by worker group and type of travel. Low-income workers make 5% more trips on average than high-income workers. Consumers also make more trips on the weekend than weekdays. Figure A11 presents the average number of daily trips by worker group and weekday vs weekend travel.

4.5 The Downtown Line and Housing Prices

The precise locations of stations on the Downtown Line (DTL) were suddenly announced on July 15, 2008. We show that the announcement coincided with large changes in residential prices for houses near each DTL station. Our empirical design is evocative of spatial difference-in-differences.[18] We compare "treated" apartments within one kilometer of a DTL station, relative to "control" apartments between one and five kilometers from one. Treated apartments are more likely (than control apartments) to directly benefit from access by foot to a DTL station, in addition to the entry of non-tradable services in the vicinity of the station.[19]

Over 4 years, relative to apartments in the outer ring, residential prices increase by 4.84% for apartments in the inner ring. See Figure 4a. Moreover, we also estimate the relationship, over time, between housing prices and distance from the DTL. The estimated regression coefficients are plotted in Figure 4b. Post-announcement, the closer the apartments are to the DTL, the higher their prices, all else equal.

Increased property prices suggest that residents highly value improvements in access from the DTL.[20] These price increases could reflect both improved access to work and consumption locations. We find that these price effects increase over time, suggesting a longer-run re-organization of economic activity.

4.6 Impact of the Downtown Line on Travel

When Stage 2 of the Downtown Line (DTL) opened, we saw a sharp uptick in travel to DTL subzones. Figure 5 plots the daily volume of trips to or from subzones with and with-

[18]For details on the precise specification and analysis, see Online Appendix B.1.

[19]To interpret our subsequent results as causal, one would have to believe that prices of treated and control apartments would have evolved similarly in the absence of the DTL. Any confounding factor driving housing prices must differentially affect apartments close to and farther from a DTL station. While an inspection of Figure 4a suggests that the parallel trends assumption holds, a large section of the DTL was built to relieve congestion along Bukit Timah Road and Dunearn Road, along which the line runs.

[20]We note that much of the housing stock (80%) in Singapore is government housing (see e.g. Wong 2013, Ferdowsian, Lee, and Yap 2023 and Lee, Ferdowsian, and Yap 2023), bought and sold on long leases. Regulation in this market is intentionally kept light-touch to facilitate price discovery and asset appreciation for existing owner-occupiers (see, e.g., Chua 2017).

out a DTL station between December 2015 and February 2016 by low- and high-income workers, with Phase 2 of the DTL opening on December 27th 2015. We observe about a 20% increase in trips after the opening. Over all subzones, the volume of low-income trips increases by a smaller percentage than high-income trips. This finding suggests that high-income workers are taking greater advantage of the new DTL than low-income workers.

We also find that low- and high-income workers adjust their travel patterns differently across neighborhoods. In Figure 6a and 6b, we plot the travel patterns by income group for two of the major locations on the DTL line separately, Bukit Panjang and Upper Bukit Timah. High-income workers increase their travel to and from Bukit Panjang by about 40% on average, while low-income workers increase their travel volume by about 20%. On the other hand, both worker groups increase their travel to and from Upper Bukit Timah by the same amount. This disparity suggests that our model should account for differential travel response in evaluating welfare across worker groups.

Lastly, we plot changes in travel patterns over a longer time horizon in Figure 7. We find that the responses in travel flow are grow over time, stabilizing about a year after the line opened. Therefore, it seems important for us to account for the long-run reorganization of economic activity in Singapore.

4.7 DTL and Non-Tradable Production

Using two cross-sections of food establishment license data between 2015 (before the opening of Phase 2 and 3 of the DTL) and 2018 (after the opening), we find that, in subzones with a DTL station, relatively more food establishments entered between 2015 and 2018. The number of food establishments in DTL locations increased by about 15% relative to non-DTL locations. See Appendix Table A3.

5 Quantitative Model

We model a closed city with heterogenous workers and travel choice over workplace and consumption locations. The city has $n \in \mathbb{N}$ neighborhoods.[21] Neighborhoods differ in their exogenous amenities, productivities, residential and commercial floor space, as well as bilateral commute times.

[21] The population of each worker group is determined exogenously. Singapore is a small city-state with tightly regulated in- and out-migration.

High- and low-income workers decide where to live, where to consume non-tradable goods and services, and where and in which sector to work.[22] Each worker type has different preferences, productivities, wages, travel costs, and consumption shares. Utility is derived from the consumption of a tradable good, a non-tradable good, and residential floor space.

On the production side, there are two sectors $G \in \{T, N\}$: tradable and non-tradable goods. Firms are located across the city and take labor and commercial floorspace as inputs. In each sector, firms differ in how much they demand different worker types: The non-tradable sector relies more on low-income workers, while the tradable sector relies more on high-income workers. Demand for non-tradable goods depends on where consumers choose to travel to consume them; whereas tradable goods can be costlessly traded across the city. Across the city, demand for labor by worker type varies, based on the productivity of each sector in each location, commercial rents, and demand for production. Perfectly competitive developers supply floor space using land and capital with constant returns to scale technology. In equilibrium, wages and the prices of floor space and goods adjust to clear the labor, land, and goods markets.

5.1 Workers

The city is populated by different worker groups indexed by $\theta \in \{+, -\}$, denoting high- and low-income workers respectively. Each type of worker has a fixed population R^θ. A worker ω in group θ chooses a location n in which to live, a location i and sector G in which to work, and a location j in which to consume non-tradable goods and services. We assume indirect utility is Cobb-Douglas and takes the form:

$$u^\theta_{niGj}(\omega) = \frac{B^\theta_n b^\theta_n(\omega)}{Q_n^{\alpha^{H\theta}} P_n^{T^{\alpha^{T\theta}}}} \times \frac{w^\theta_{iG} a^\theta_{iG}(\omega)}{\exp(\kappa^\theta \tau_{ni})} \times \frac{s^\theta_j(\omega)}{P_j^{N^{\alpha^{N\theta}}} \exp(\kappa^\theta \tau_{nj})}, \tag{1}$$

where $0 < \alpha^{T,\theta}, \alpha^{H,\theta}, \alpha^{N,\theta} < 1$ and $\alpha^{T,\theta} + \alpha^{H,\theta} + \alpha^{N,\theta} = 1$. To ease exposition in this subsection, fix a residential neighborhood n, work neighborhood i, sector G, consumption neighborhood j and worker type θ.

Individuals consume residential floorspace, a tradable good, and a non-tradable good. We accommodate different preferences, productivities, wages, travel costs and consumption shares by type. The first term in the product in Equation (1) covers the utility factors

[22] In additional analyses, we allow each worker to make many consumption trips to different locations. We find that this extension collapses to the same estimating equations as in the main model. This material is available in a supplement provided by the authors upon request.

specific to the worker's residence. She receives B_n^θ in common residential amenities in neighborhood n. Differences in residential preferences across groups will drive worker types to live in different neighborhoods. However, this worker dislikes more expensive residential floor space Q_n and tradable goods P_n^T. We assume that the tradable good is traded without cost in the city and assume it is our numeraire: $P_n^T = P^T = 1$.

The second and third terms in the product in Equation (1) correspond to the utility terms specific to the worker's workplace and consumption neighborhood respectively. She earns a wage w_{iG}^θ per efficiency unit in neighborhood i. Meanwhile, she dislikes a more expensive price P_j^N of the non-tradable good in neighborhood j. To work in neighborhood i and consume non-tradables in neighborhood j, the worker pays iceberg travel costs $\exp(\kappa^\theta \tau_{ni})$ and $\exp(\kappa^\theta \tau_{nj})$ respectively. These travel costs increase with the time τ it takes to travel between neighborhoods. The parameter κ^θ controls the size of these travel costs by worker group. We expect differences in travel costs since low-income workers receive subsidies on travel. Higher travel costs imply larger gains from reductions in travel time.

Next, high- and low-income workers have different consumption shares over housing, $\alpha^{H,\theta}$, tradables, $\alpha^{T,\theta}$, and non-tradables, $\alpha^{N,\theta}$. These shares measure the relative importance of housing, tradable and non-tradable consumption for utility for people of type θ. Larger consumption shares on housing imply larger implications of changes in rents for welfare. Larger consumption shares on non-tradables imply larger implications of access to consumption opportunities for welfare.

Finally, worker ω has idiosyncratic draws for residential amenities in neighborhood n, $b_n^\theta(\omega)$, efficiency units of labor (productivity) in neighborhood i and sector G, $a_{iG}^\theta(\omega)$, and consumption amenities in neighborhood j, $s_j^\theta(\omega)$). We assume that these idiosyncratic shocks are drawn from the following independent Frechet distributions:

$$\begin{aligned} F_n^{R\theta}(b) &= \exp(-T_n^{R\theta} b^{-\epsilon^{R\theta}}) \\ F_{iG}^{L\theta}(a) &= \exp(-T_i^{L_I\theta} T_G^{L_G\theta} a^{-\epsilon^{L\theta}}) \\ F_j^{S\theta}(s) &= \exp(-T_j^{S\theta} s^{-\epsilon^{S\theta}}) \end{aligned} \tag{2}$$

where all the shape parameters, ϵ, are greater than 1; and all the scale parameters, T, are greater than zero. The scale parameters, $T_n^{R\theta}, T_i^{L_I\theta}, T_G^{L_G\theta}$, and $T_j^{S\theta}$, control the overall level of the draws for residential preferences, work location productivity, sector productivity, and consumption preferences respectively. We allow scale parameters to vary across worker groups, to capture differences in preferences and productivities over locations across types. We also allow shape parameters, $\epsilon^{R\theta}, \epsilon^{L\theta}, \epsilon^{S\theta}$, which control the dis-

persion of the distributions to differ across groups. A higher ϵ corresponds to a smaller dispersion of idiosyncratic taste shocks. The sensitivity of choices to other variables such as travel costs is governed by the dispersion of preferences or productivity. When workers have similar matches in different locations (high ϵ), their choices are more sensitive to these other variables. Differences in heterogeneity across groups will be important in determining the incidence of travel costs, since these differences control the extent to which individuals are willing to bear high travel costs to work or consume in a location.

Timing. Workers first choose where to live; then, in a second step, they choose where to work and where to consume non-tradables.[23] First, individuals observe their idiosyncratic residential amenities draws for all neighborhoods and choose to live in some neighborhood n. Second, individuals observe their idiosyncratic productivities for all workplaces and idiosyncratic consumption preferences in all neighborhoods, then choose to work in some neighborhood i and sector G, and to consume non-tradables in some neighborhood j. We solve the worker problem by backward induction.

5.1.1 Employment Location Decision

Having chosen a neighborhood n in which to live, individuals draw a vector of match-productivities with firms across the city as in Equation (2). With these draws in hand, workers choose to work in the neighborhood that offers the highest income net of their type-specific commuting costs:

$$\max_{i,G} \frac{w_{iG}^{\theta} a_{iG}^{\theta}(\omega)}{\exp(\kappa^{\theta}\tau_{ni})}. \tag{3}$$

Fix a worker type θ who has made the choice to live in neighborhood n. Since idiosyncratic productivities are distributed according to a Frechet distribution, the probability she decides to work in neighborhood i and sector G is given by the following gravity equation:

$$\lambda_{niG|n}^{L\theta} = \frac{T_i^{L_I\theta} T_G^{L_G\theta} [w_{iG}^{\theta} \exp(-\kappa^{\theta}\tau_{ni})]^{\epsilon^{L\theta}}}{\sum_l \sum_H T_l^{L_I\theta} T_H^{L_G\theta} [w_{lH}^{\theta} \exp(-\kappa^{\theta}\tau_{nl})]^{\epsilon^{L\theta}}} \tag{4}$$

Individuals are more likely to travel to work in a neighborhood that pays a high wage net of travel costs, as in the numerator, relative to those in all other locations, as in the denominator. The sensitivity of employment decisions to commute costs is governed by the dispersion of productivities across neighborhoods. Similar matches across workplaces

[23] An alternative specification could be to first allow workers to choose where to work, then where to live, then where to consume. If our mainline model is correct, this alternate specification will deliver the same quantitative results. However, in other settings in which trip chaining is of first order concern, switching the order of worker choice could result in different implications for estimation and welfare.

in different locations and sectors (high ϵ) imply that choices are more sensitive to commute costs. Differences in productivity heterogeneity across worker types is important in determining the incidence of travel costs, controlling the extent to which workers are willing to bear high commute costs to work in a neighborhood. Differences in productivity across neighborhoods and commute costs by type drive differences in work travel patterns across worker groups.

Expected income prior to drawing the vector of match productivities is directly related to the denominator in Equation (4) through

$$\mathbb{W}_n^\theta \triangleq \mathbb{E}_n^{L\theta}[v] = \Gamma\left(\frac{\epsilon^{L\theta}-1}{\epsilon^{L\theta}}\right)\left[\sum_l \sum_H T_l^{L_l\theta} T_H^{L_G\theta}[w_{lH}^\theta \exp(-\kappa^\theta \tau_{nl})]^{\epsilon^{L\theta}}\right]^{1/\epsilon^{L\theta}} \tag{5}$$

where $\Gamma(.)$ is the Gamma function. Intuitively, in locations with better *access to employment* $\mathbb{W}_n^\theta$, or access to locations with high expected income by type, workers are better off.

5.1.2 Consumption Location Decision

Having chosen to live in neighborhood n, workers draw a vector of idiosyncratic preference shocks with consumption locations across the city as in Equation (2). With these draws in hand, workers choose to consume non-tradables in the neighborhood with the best consumption amenities, accounting for travel costs and the price of non-tradables.

$$\max_j \frac{s_j^\theta(\omega)}{P_j^{N^{\alpha^N}} \exp(\kappa \tau_{nj})} \tag{6}$$

Again, fix a worker type θ and residence location n. Since idiosyncratic consumption tastes $s_j^\theta(\omega)$ are distributed according to a Frechet distribution, the probability this worker decides to consume non-tradables in j is given by the following gravity equation:

$$\lambda_{nj|n}^{S\theta} = \frac{T_j^{S\theta}[P_j^{N^{\alpha^N}} \exp(\kappa \tau_{nj})]^{-\epsilon^{S\theta}}}{\sum_m T_m^{S\theta}[P_m^{N^{\alpha^N}} \exp(\kappa^\theta \tau_{nm})]^{-\epsilon^{S\theta}}} \tag{7}$$

Individuals are more likely to travel to consume in neighborhoods with high consumption amenities net of the price of non-tradables and travel costs, as in the numerator, relative to those in all other locations, as in the denominator. The sensitivity of consumption decisions to travel costs is governed by the dispersion of preferences across neighborhoods. Similar amenities across consumption locations (high ϵ) imply that choices

are more sensitive to travel costs. Differences in preference heterogeneity across worker types is important in determining the incidence of travel costs, controlling the extent to which workers are willing to bear high travel costs to consume non-tradables in a neighborhood. Differences in preferences across neighborhoods and travel costs by type drive differences in consumption travel patterns across worker groups.

Expected utility from non-tradable consumption prior to drawing the vector of idiosyncratic preferences is directly related to the denominator in Equation (7) through

$$\mathbb{S}_n^\theta \triangleq \mathbb{E}_n^{S\theta}[\gamma] = \Gamma\left(\frac{\epsilon^{S\theta}-1}{\epsilon^{S\theta}}\right)\left[\sum_m T_m^{S\theta}[P_m^{N^{\alpha^N}}\exp(\kappa^\theta\tau_{nm})]^{-\epsilon^{S\theta}}\right]^{\alpha^{N\theta}/\epsilon^{S\theta}} \tag{8}$$

where $\Gamma(.)$ is the Gamma function. Intuitively, in locations with better *access to consumption* $\mathbb{S}_n^\theta$, or access to locations with low prices and high amenities by type, workers are better off.

5.1.3 Residential Location Decision

In the first stage, individuals choose where to live to maximize their expected indirect utility after observing their idiosyncratic residential amenity draws across all neighborhoods. Workers of type θ solve the following problem:

$$\max_n U_n^\theta(\omega) = \frac{B_n^\theta b_n^\theta(\omega)}{Q_n^{\alpha^H}}\mathbb{W}_n^\theta\mathbb{S}_n^\theta \tag{9}$$

Workers are attracted to locations with high residential amenities, low housing prices, high net incomes, and high utility from consumption of non-tradables.

Since idiosyncratic residence amenity draws are Frechet distributed, the share of type-θ workers who live in neighborhood n is

$$\lambda_n^{R\theta} = \frac{T_n^{R\theta}[P_n^{T^{-\alpha^T}}Q_n^{-\alpha^H}B_{n\theta}\mathbb{W}_{n\theta}\mathbb{S}_{n\theta}]^{\epsilon^{R\theta}}}{\sum_k T_k^{R\theta}[P_k^{T^{-\alpha^T}}Q_k^{-\alpha^H}B_{k\theta}\mathbb{W}_{k\theta}\mathbb{S}_{k\theta}]^{\epsilon^{R\theta}}} \tag{10}$$

Similar residential amenities across neighborhoods (high ϵ) imply that choices are more sensitive to changes in access to employment, access to consumption, and rents. Differences in preferences across neighborhoods by type drive differences in residential patterns in the city across worker groups.

5.1.4 Expected Utility

Since we consider a "closed city", expected utility from living in the city prior to drawing the vector of idiosyncratic preferences is directly related to the denominator in Equation (10) through

$$\bar{U}^{\theta} = \Gamma\left(\frac{\epsilon^{R\theta}-1}{\epsilon^{R\theta}}\right)\left[\sum_{k} T_k^{R\theta}[P_k^{T^{-\alpha^T}} Q_k^{-\alpha^H} B_{k\theta} \mathbb{W}_{k\theta} \mathbb{S}_{k\theta}]^{\epsilon^{R\theta}}\right]^{1/\epsilon^{R\theta}}. \tag{11}$$

5.2 Firms

5.2.1 Technology

In each location $i \in \mathbb{N}$, firms operate in both tradable and non-tradable sectors, $G \in \{N, T\}$. Sectoral productivity differs across locations. Firms produce under perfect competition and with a constant returns to scale technology. We specify firm production as being a Cobb-Douglas aggregate over labor and commercial floor space. Output is thus

$$Y_{iG} = A_{iG} L_{iG}^{\beta^G} H_{iG}^{1-\beta^G} \tag{12}$$

where $\beta^G \in [0,1]$ and H_{iG} is commercial floor space. Meanwhile, labor input L_{iG} is a Cobb-Douglas aggregate over each worker group's effective units of labor, $\tilde{N}_{iG\theta}$:

$$L_{iG} = \prod_{\theta} \tilde{N}_{iG\theta}^{\beta^{G\theta}} \tag{13}$$

where $\sum_{\theta} \beta^{G\theta} = 1$. Each sector and neighborhood has different productivities A_{iG}. Sectors differ in the intensity in which they use different types of workers $\beta^{G\theta}$. The tradable sector requires a higher share of high-income workers, while the non-tradable sector heavily relies on low-income workers. Thus, the spatial distribution of non-tradable versus tradable production is strongly related to that of low-income versus high-income jobs. As a result, different worker types living in the same neighborhood may have different levels of access to employment opportunities.

5.2.2 Factor Demand

Under perfect competition and free entry, firms make zero profits in equilibrium. Furthermore, the price of each variety is equal to its marginal cost:

$$P_i^G = A_{iG}{}^{-1} W_{iG}^{\beta^G} q_i^{1-\beta^G} \tag{14}$$

where q_i is the price of commercial floor space in workplace i and

$$W_{iG} = w_{iG+}^{\beta^{G+}} w_{iG-}^{\beta^{G-}} \tag{15}$$

is the cost of labor for sector G in location i. In this definition, $w_{iG\theta}$ is the wage per efficiency unit for a worker of type θ. Wages are different across both sectors and locations, with each sector and location pair facing an upward-sloping supply function for effective units of labor for each worker type. Solving the firm's profit maximization problem, we find that the demand for labor and commercial floorspace can be written as

$$L_{iG} = \beta^G P_{iG} Y_{iG} / W_{iG} \tag{16}$$

$$H_{iG} = (1 - \beta^G) P_{iG} Y_{iG} / q_{iG} \tag{17}$$

$$\tilde{N}_{iG\theta} = \beta^{L\theta,G} L_{iG} W_{iG} / w_{iG\theta} \tag{18}$$

5.3 Land Market

Following Ahlfeldt et al. (2015), after factoring in the tax equivalent of land use regulations, the land market equilibrium requires no arbitrage between the commercial and residential use of floor space. The commercial price of floor space for both the tradable and non-tradable sector is

$$q_i = \xi_i Q_i, \tag{19}$$

where ξ_i equals one plus the tax equivalent of land use regulations that restrict commercial land use relative to residential land use. We allow this wedge between commercial and residential floor prices to vary across neighborhoods.

Floor space is supplied by perfectly competitive developers using land K_i and capital M_i, with constant returns to scale technology:

$$H_i = M_i^{\mu_H} K_i^{1-\mu_H} \tag{20}$$

where H_i is total floor space and μ_H is the share of land in floor space production. Therefore, the corresponding dual cost function for floor space is

$$Q_i = \mu_H^{-\mu_H} (1 - \mu_H)^{-(1-\mu_H)} \mathbb{P}^{\mu_H} \mathbb{R}_i^{1-\mu_H}, \tag{21}$$

where $\mathbb{P}$ is the common price for capital across all neighborhoods, and $\mathbb{R}_i$ is the price for

land in neighborhood i. Cost minimization implies that

$$Q_i = \mathbb{P} K_i^{\frac{\mu_H - 1}{\mu_H}} H_i^{\frac{1-\mu_H}{\mu_H}} \mu_H^{-1} \tag{22}$$

As the production of non-tradable goods and the demand for floor space rise in response to increased consumption travel, increased rents may drive out tradable production and residents, shifting commercial and residential spatial patterns across the city.

5.4 Market Clearing

Land. The expression for demand for residential floor space in neighborhood n is

$$H_n^R = \frac{\sum_\theta \alpha^{H,\theta} R^\theta \lambda_n^R \mathbb{W}_n}{Q_n} \tag{23}$$

where we sum over the housing expenditures for all the residents in neighborhood n across worker groups, using the fact that expenditures on residential floor space is a constant share of income (since workers have Cobb-Douglas utility).

Market clearing for floor space requires that the total supply of floor space equals the total floor space demanded from both residents and firms in each neighborhood j:

$$H_j = H_j^R + \sum_{G \in \{N,T\}} H_{jG}. \tag{24}$$

Labor. Using the commuting probabilities from Equation (4), the supply of workers to any location is the sum over the number of commuters by origin that travel to that work location:

$$N_{iG\theta} = \sum_n \lambda_{niG|n}^{L\theta} \lambda_n^R R^\theta \tag{25}$$

Labor supply in the model takes a log-linear form that depends on two forces. Firstly, more workers commute to destinations paying higher wages. Second, firms attract workers when they have better access to them through the commuting network. Ultimately, workers care about wages net of commute costs. Meanwhile, total effective labor supply to location i is given by

$$\tilde{N}_{iG\theta} = \sum_n R_n \lambda_{ni|n}^{L\theta} \lambda_{niG|ni}^{L\theta} \bar{a}_{niG}^\theta \tag{26}$$

where $\bar{a}_{niG}^\theta$ is the average productivity of type-θ workers who live in n and decide to work

in i. Using the properties of the Frechet distribution, we have[24]

$$\bar{a}^{\theta}_{niG} = \Gamma(\epsilon^{L\theta})(T_i^{L_I\theta} T_G^{L_G\theta} / \lambda^{L\theta}_{niG|n})^{1/\epsilon^{L\theta}} \tag{27}$$

In each neighborhood i and sector G, market clearing requires that the supply of effective labor, as in Equation (26), equals the demand for effective labor, as in Equation (16). Wages by worker type are endogenously determined by market clearing.

Non-tradables. In each neighborhood j, total receipts for non-tradables must equal total expenditures on non-tradables:

$$P_j^N A_{iN} L_{iN}^{\beta^N} H_{iN}^{1-\beta^N} = \sum_{n \in \mathbb{N}} \sum_{\theta} \alpha^{N,\theta} \lambda^{S\theta}_{nj|n} \lambda^R_n R^{\theta} \mathbb{W}_{n\theta}. \tag{28}$$

Prices of non-tradables are endogenously determined to clear the market. We sum over expenditures of non-tradables for all the workers who travel from some neighborhood n to consume in neighborhood j across worker groups, using the fact that expenditures on non-tradables is a constant share of income (since worker utility is Cobb-Douglas).

5.5 Externalities

5.5.1 Productivities

A location's productivity depends on both an exogenous component $\bar{A_{iG}}$ that reflects features independent of economic activity (e.g. access to roads, slope of land) as well as the endogenous density of employment in that location:

$$A_{iG} = \bar{A}_{iG} (\sum_{\theta} N_{i\theta} / \bar{H}_i)^{\mu_A} \tag{29}$$

where $\bar{H}_i$ is the total units of land in location i. The strength of agglomeration externalities is governed by the parameter μ_A.[25]

[24] $\bar{a}^{\theta}_{niG} = \mathbb{E}[a^{\theta}_{inG} | T_i^{L_I\theta} T_G^{L_G\theta} (w^{\theta}_{iG} \exp(-\kappa\tau_{ni}))^{\epsilon^{L\theta}} > T_l^{L_I\theta} T_H^{L_G\theta} (w^{\theta}_{lH} \exp(-\kappa\tau_{nl}))^{\epsilon^{L\theta}}, \quad \forall l, H]$

[25] While we permit spillovers in productivity by sector, our specification here disallows these spillovers to be type-specific. We collapse productivity spillovers by type for two reasons. The first is parsimony. The second is that, absent spillovers of any kind, our model generates qualitatively similar counterfactual predictions. See Table A7 in the Appendix. Were we to include type-specific productivity spillovers, we would expect workers of the same (income, sector) bin to concentrate in neighborhoods in which they are already abundant.

5.5.2 Amenities

Amenities in a neighborhood are a function of an exogenous component $\bar{B}_n$ and a residential externality. In turn, this externality depends on the endogenous ratio of high-skilled and low-skilled residents:

$$B_n^\theta = \bar{B}_n^\theta (R_n^\theta / R_n^{-\theta})^{\mu_{U\theta}} \tag{30}$$

In our model, instead of being a function of the total density of residents, endogenous amenities depend on the demographic composition of workers across income groups, similar to Tsivanidis (2019). Workers are more willing to pay to live in neighborhoods that are high in amenities of their type. As workers locate in these neighborhoods, they increase these type-specific endogenous amenities even more, strengthening the segregation effect. We let the strength of residential externalities differ across worker groups and estimate the relative strength of these forces.

5.6 Equilibrium

We now define the general equilibrium of our model.

Definition Given vectors of exogenous location characteristics $\{\bar{B}_n^\theta, \bar{A}_{iG}, \tau_{ij}, \bar{H}_j, \psi_i, \xi_i\}$, city group-wise populations $\{R^\theta\}$, and model parameters $\{\alpha, \beta, \kappa^\theta, T_n^{R\theta}, T_i^{L_I\theta}, T_G^{L_G\theta}, T_j^{S\theta}, \epsilon^{R\theta}, \epsilon^{L\theta}, \epsilon^{S\theta}, \mu_H, \mu_A, \mu_{U\theta}\}$, an equilibrium is defined as a vector of endogenous objects $\{q_i, w_{iG}^\theta, H_{iG}, \tilde{N}_{iG\theta}, \lambda_n^{R\theta}, \lambda_{niG|n}^{L\theta}, \lambda_{nj|n}^{S\theta}, \bar{U}^\theta\}$ such that:

1. **Labor Market Clearing**: The supply of labor by individuals in Equation (26) is consistent with demand for labor by firms in Equation (16).

2. **Floorspace Market Clearing**: The market for floorspace clears as in Equation (24), its price is consistent with residential populations (23) and commercial floorspace use (17), and total floor space is consistent with land developer optimality (22).

3. **Goods Market Clearing**: Non-tradable consumption matches non-tradable production in each neighborhood, as in Equation (28).

4. **Closed City**: Populations add up to the city total.

We discuss equilibrium existence and uniqueness. In a version of the model without externalities, by similar arguments to those made in Ahlfeldt et al. (2015), the model's congestion forces — commuting costs, travel costs to consume non-traded goods, and an inelastic land supply — ensure that a unique equilibrium exists. In the full version of the model, by adapting arguments in Section 3.1 of Allen, Arkolakis, and Li (2022),

uniqueness can be guaranteed when "agglomeration forces are small relative to congestion forces."[26]

5.7 Model Discussion

First of all, our model captures the direct distributional implications of changing access to consumption, in addition to changing employment access. Transit improvements affect how easily different worker types access consumption opportunities, and these effects also depend on where they (want to) live, work and consume. Suppose travel costs fell between low-income neighborhoods and locations in which non-tradable goods are cheap and high in (type-specific) quality. Then low-income workers would become better off. Access to consumption is more important to welfare for worker types with 1) larger expenditures on non-tradables, 2) larger dispersion of preferences over consumption locations (more inelastic demand), and 3) higher travel costs.

Next, consumption travel shapes the residential patterns of the city in a type-specific manner. Workers trade off access to non-tradables, along with access to employment and higher rents when choosing where to live. Worker types with less dispersion in residential amenities are better able to take advantage of both improved access to consumption and employment in other locations. In contrast with that of Miyauchi, Nakajima, and S. Redding (2022), our specification of consumption travel preserves tractability while still allowing us to speak to differences in behavior between low- and high-income workers.

Lastly, incorporating travel to consume non-tradables also has distributional implications for access to employment across worker groups. Responding to changes in transit access, consumers change where they consume non-tradables, trading off travel costs and type-specific amenities and prices. Thus relative factor demand changes across locations. Since firms selling non-tradables hire more low-income workers, low-income jobs move to where residents are traveling more for consumption (thus demanding more non-tradable production in those locations). High-income jobs in tradables may also move away from locations with greater consumption travel, since commercial rents will rise in response to greater production of non-tradables. Therefore, low-income jobs could move

[26]Characterizing formal sufficient conditions for uniqueness involves tedious algebra. We sketch the approach here. First, bound the absolute values of the elasticities of productivity spillovers with respect to worker population; and those of amenity spillovers with respect to residential population. Next, substitute the gravity and spillover equations into the market clearing equations in residential population, labor supply, and consumption travel. This substitution gives rise to bounds on the elasticities of populations in each workplace, residence, and consumption location by type. Collect these elasticity bounds in a matrix of "cross-elasticities." Then, by Remark 1 and part (i) of Theorem 1 of Allen, Arkolakis, and Li (2022), for equilibrium uniqueness, it is sufficient that the spectral density of the matrix of "cross-elasticities" be less than one.

to locations closer or further from where low-income workers live, locations with higher or lower wages for low-income workers, or locations which are more or less productive for low-income workers, and analogously for high-income workers. Since commercial activity between tradable and non-tradable sectors reorganizes across space, each worker group faces differential changes in expected access to employment or expected income net of commuting. Therefore, it is an empirical question whether inequality rises or falls.[27]

6 Estimation

This section estimates the model from Section 5.

First, using gravity specifications from the model, we estimate workplace and consumption preferences and access across neighborhoods and across worker groups. In this first step, we exploit two sources of variation. One source is cross-sectional: we see variation in travel times and variation in commuting (or consumption travel) flows between residence-destination pairs by income type. The other source of variation is generated by the opening of DTL2: differences between flows shortly before and after the line was opened identify the impact of lower travel times on commuting (or consumption travel) choices. Meanwhile, differences between flows in the long run before and after DTL2 was opened identify changes in the spatial (re-)organization of economic activity.

Second, we estimate type-specific work and consumption travel costs, as well as parameters governing the dispersion of idiosyncratic workplace productivities and consumption amenities. For workplace productivity, by combining variation in residential incomes by type across neighborhood with variation in the estimated residence fixed effects in the first step, we can identify the dispersion of idiosyncratic workplace productivities by type, $\hat{\epsilon}^{L\theta}$. Using the estimated values of $\hat{\epsilon}^{L\theta}$ and the semi-elasticities for consumption and work travel, we can recover the dispersion of idiosyncratic consumption amenities by worker type and the travel costs per worker type.

Third, we estimate parameters that govern the dispersion of idiosyncratic residential amenities and residential spillovers by type. When DTL2 opened, travel times fell. In the long run, workers and firms re-optimize, shifting the spatial distribution of economic activity in Singapore. Therefore, variation in residence shares, rents, market access, and residential composition across time identifies amenity spillovers across space.

Fourth, we recover residential amenities across neighborhoods and across high- and low-income workers. We validate that the model-implied residential amenities are highly

[27] These reallocative predictions of the model could be directly tested with data on exogenous shocks to demand for non-tradable goods.

correlated with external data, including the spatial distribution of schools, community clubs, and parks.

Fifth, we show how the model can be inverted to obtain unobservable endogenous variables that rationalize the observed data as an equilibrium of the model.[28]

6.1 Estimating Consumption Preferences and Access

First, we estimate workplace and consumption preferences and access using data from before the DTL opened. For each worker type $\theta \in \{+, -\}$, we combine the consumption travel flow Equation (7) with the specification of travel costs $\phi^{S\theta} = \epsilon^{S\theta}\kappa^{\theta}$. The resulting gravity equation relates the variation in consumption travel flows to the variation in travel times:

$$\ln \lambda^{S\theta}_{nj|n} = \eta^{S\theta}_{j} - \mu^{S\theta}_{n} - \phi^{S\theta}\tau_{nj} + u^{S\theta}_{nj}. \tag{31}$$

In this equation, $\eta^{S\theta}_{j}$ and $\mu^{S\theta}_{n}$ are workplace and origin fixed effects, and $u^{S\theta}_{ni}$ is an unobserved shifter of consumption travel propensities. We take τ_{nj} as the average time taken to travel between neighborhoods n and j in the farecard data, and construct work travel shares from identified residences and consumption trips in the farecard data.

We note that travel times τ may be correlated with the unobservable $u^{S\theta}_{nj}$. Endogeneity in this setting takes two forms. The first is conventional: transit development may be driven precisely by where workers travel to consume non-tradables. To deal with this source of endogeneity, in our preferred specification, we instrument for τ with the straight-line distance between neighborhood centroids. Columns 3 and 4 of Table 1 report the results from the regression associated with Equation (31). We estimate a negative and statistically significant semi-elasticity of consumption travel flows with respect to travel time. These semi-elasticities take the value of $\hat{\phi}^{S+} = -0.053$ for high-income workers and $\hat{\phi}^{S-} = -0.094$ for low-income workers. Thus, we find that low-income workers are more sensitive to changes in travel time. In the next subsection, we will show that, across groups, their respective gravity coefficients are larger for consumption travel than work travel.

The second source of endogeneity is from possible sorting and complementarities across (origin, consumption destination) pairs (see e.g. Dingel and Tintelnot 2020 and Severen 2019). To assess robustness of our preferred specification to such complementarities, we also estimate Equation (31) using a difference-in-differences framework:

[28] We demonstrate the invertability of this model even though our counterfactual analysis employs "exact-hat algebra."

$$\Delta \ln \lambda_{nj|n}^{S\theta} = \Delta\eta_j^{S\theta} - \Delta\mu_n^{S\theta} - \phi^{S\theta}\Delta\tau_{nj} + \Delta u_{nj}^{S\theta}, \tag{32}$$

where the differences are across trips taken before and after the line opening. Identification of $\phi^{S\theta}$ then requires $\mathbb{E}[\Delta\tau_{nj} \times \Delta u_{nj}^{S\theta}] = 0$, i.e. that sorting patterns and flow correlations across distance are stable.

Instead of pooling across time, this difference-in-differences method enables us to leverage two "intermediate" control groups of origin-destination *pairs*, where one of the two "ends" is treated and the other is not. Consider workers living in Bishan, an origin neighborhood not sited on top of a DTL2 station. Workers consuming non-tradable goods in Beauty World (who benefit from DTL2) are compared with those who visit Orchard (who do not).[29] Similarly, consider Orchard, a major shopping belt not served directly by DTL2. Shoppers who live in Bishan (not on DTL2) are compared with those who live in Bukit Panjang (a terminus of DTL2). These comparisons "control for many potential unobserved motives for changing travel behavior" (Severen 2019).

The exclusion restriction would be violated if, for instance, after DTL2 opened, shops in all neighborhoods served by DTL2 systematically favored demographic groups prevalent in the far west of Singapore (hence closer to DTL2 neighborhoods) relative to those in the east. Such a violation seems implausible because Singapore is highly racially and socioeconomically integrated.

In columns 3 and 4 of Table A5, we display the results from the regression associated with Equation (32). We find that the semi-elasticities recovered in this alternate procedure are quantitatively similar to those recovered when estimating the first equation (in levels).

The consumption location fixed effect $\eta_j^{S\theta}$ from the first regression gives us measures of consumption attractiveness across neighborhoods:

$$\exp(\hat{\eta}_j^{S\theta}) = T_j^{S\theta} P_{jN}^{-\alpha^N} \epsilon^{S\theta}, \tag{33}$$

which captures average consumption amenities by type, as well as the prices of non-tradables in each neighborhood i. In Figure 8a, we display a scatter plot of consumption preferences by high- and low-income workers. We observe substantial differences in where low and high income groups prefer to consume non-tradables. As a test of model fit, in Table 2, for each income group, we display regression coefficients of consumption attractiveness against various external measures of retail amenities. We find that our attractiveness measure is positively correlated with the density of food establishments, su-

[29]The fastest transit route from Bishan to Beauty World is to take the Circle Line from Bishan to the Botanic Gardens, then transfer to the Downtown Line headed northwest to Beauty World. In contrast, Orchard is accessible from Bishan via the North-South Line, sans transfers.

permarkets and clinics in a neighborhood. Interestingly, we find that low-income workers prefer locations where low-cost options make up a larger share of food establishments, such as hawker stalls and food courts, while the corresponding coefficient is insignificant at the 5% level for high-income workers. This last finding corroborates our descriptive evidence that the rich and poor consume different non-tradable goods.

The residence fixed effect term $\mu_n^{S\theta}$ from the above regression gives us

$$\exp(\hat{\mu}_n^{S\theta}) = \sum_{m \in \mathbb{N}} T_m^{S\theta} P_m^{N^{-\alpha^N \epsilon^{S\theta}}} \exp(-\epsilon^{S\theta} \kappa^\theta \tau_{nm}) \tag{34}$$

which, for fixed $\epsilon^{S\theta}$ and origin neighborhood n, allows us to recover our measure of consumption access, $\mathbb{S}_n^\theta$, across neighborhoods. This measure reflects the expected utility from non-tradable consumption across destination neighborhoods j.

6.2 Estimating Workplace Preferences and Access

Next, we estimate workplace preferences and access. For each worker type $\theta \in \{+, -\}$, we combine the work travel flow Equation (4) with travel costs $\phi^{L\theta} = \epsilon^{L\theta}\kappa^\theta$. The resulting gravity equation relates the variation in work travel flows to the variation in travel times:

$$\ln \lambda_{ni|n}^{L\theta} = \eta_i^{L\theta} - \mu_n^{L\theta} - \phi^{L\theta}\tau_{ni} + u_{ni}^{L\theta} \tag{35}$$

where $\eta_i^{L\theta}$ and $\mu_n^{L\theta}$ are fixed effects, and $u_{ni}^{L\theta}$ is an unobserved shifter of work travel propensities. As before, we let τ be the average travel time between neighborhoods in the farecard data, and construct work travel shares from identified residences and workplaces in the farecard data.

As for the consumption gravity equation, travel times τ_{ni} may be correlated with the unobservable $u_{ni}^{L\theta}$. Again, we take two approaches to deal with endogeneity. First, travel times may be correlated with unobserved variation in transit development that affects workplace choice. To deal with this issue, we instrument for τ with the straight-line distance between neighborhood centroids. Columns 1 and 2 of Table 1 reports the results from estimating Equation (35). We estimate a negative and statistically significant semi-elasticity of work travel flows with respect to travel time of $\hat{\phi}^{L+} = -0.051$ for high-income workers and $\hat{\phi}^{L-} = -0.070$ for low-income workers. Again, we find a gravity coefficient for low-income workers that is larger in magnitude, indicating that low-income workers are more sensitive to changes in travel time. Comparing our estimates for work trips to those of consumption trips, we find smaller gravity coefficients for work travel than consumption travel across both groups.

As before, a second source of endogeneity is from possible sorting and complementarities across (origin, workplace destination) pairs. To assess robustness of our preferred specification to such complementarities, we also estimate Equation (36) using a difference-in-differences framework:

$$\Delta \ln \lambda_{ni|n}^{L\theta} = \Delta\eta_i^{L\theta} - \Delta\mu_n^{L\theta} - \phi^{L\theta}\Delta\tau_{ni} + \Delta u_{ni}^{L\theta}, \tag{36}$$

where the differences are taken across trips taken before and after the line opening. Identification of $\phi^{L\theta}$ then requires $\mathbb{E}[\Delta\tau_{ni} \times \Delta u_{ni}^{L\theta}] = 0$, i.e. that sorting patterns and flow correlations across distance are stable. Similar arguments as in the previous subsection apply. The exclusion restriction is violated if, for instance, after DTL2 opened, workplaces in all neighborhoods served by DTL2 systematically favored demographic groups prevalent in the far west of Singapore relative to those in the east. Such a violation seems implausible because of laws against discrimination in hiring, in addition to the fact that Singapore is highly racially and socioeconomically integrated.

In columns 1 and 2 of Table A5, we display the results from the regression associated with Equation (36). Again, we find that the semi-elasticities recovered in this alternate procedure are quantitatively similar to those recovered when estimating the first equation (in levels).

The workplace fixed effect from estimation gives us measures of workplace attractiveness across neighborhoods:

$$\exp(\hat{\eta}_i^{L\theta}) = \sum_G T_i^{L_I\theta} T_G^{L_G\theta} w_{iG}^{\epsilon^{L\theta}}, \tag{37}$$

which captures type-specific averages, across sectors, of productivity and wages in each neighborhood i. Figure 8b presents a scatter plot of workplace preferences by high- and low-income workers. We observe substantial differences in where low and high income groups prefer to work. In Table 3a, we find that workplace attractiveness for both groups is highly correlated with employment density.

The residence fixed effect from estimation gives us

$$\exp(\hat{\mu}_n^{L\theta}) = \sum_{l\in\mathbb{N}} \sum_G T_i^{L_I\theta} T_G^{L_G\theta} w_{lG}^{\epsilon^{L\theta}} \exp(-\epsilon^{L\theta}\kappa^\theta\tau_{nl}). \tag{38}$$

Given idiosyncratic shocks $\epsilon^{L\theta}$ and an origin neighborhood n, we can recover our measure of employment access, $\mathbb{W}_n^\theta$. This measure captures the expected wages net of commuting across neighborhoods i and sectors G.

6.3 Parameter Estimation: Travel Elasticities and Costs

6.3.1 Dispersion of Workplace Idiosyncratic Productivity

Taking logs of the expression for expected income by worker type in Equation (5), we can estimate the workplace dispersion parameter, $\epsilon^{L\theta}$, by regressing the estimated residence fixed effects, $\hat{\mu}_n^{L\theta}$, on the log of average residential income from observed administrative data by type. The parameter $\epsilon^{L\theta}$ is thus identified off variation across neighborhoods in income and the estimated measure of residence attractiveness.[30]

$$\hat{\mu}_n^{L\theta} = \epsilon^{L\theta} \log(\text{Residential Income}_n) + e_n \tag{39}$$

Table 3b presents our estimation results. We find a strong positive correlation between our model's estimate of expected income with that of non-targeted data. We show in Figure 9, similar to what Miyauchi, Nakajima, and S. Redding (2022) find, that residence fixed effects are generally log-linear in residential income. However, there are outliers in a few high-income neighborhoods, reflecting non-labor income at high income levels. In our preferred specification, we drop the four richest neighborhoods, estimating a workplace idiosyncratic dispersion of $\epsilon^{L+} = 2.912$ for high-income workers and $\epsilon^{L-} = 5.023$ for low-income workers respectively. High-income workers have more inelastic work travel patterns than low-income workers. These findings are consistent with those in Tsivanidis (2019). Thus, high-income workers benefit more from reductions in travel costs as they are less able to substitute to more attractive work locations in equilibrium.

6.3.2 Dispersion of Idiosyncratic Consumption Amenities

Next, we can compute the consumption travel dispersion parameter by worker type, $\epsilon^{S\theta}$, using the estimated semi-elasticities for consumption and work travel, $\phi^{S\theta}$ and $\phi^{L\theta}$:

$$\hat{\epsilon}^{S\theta} = \hat{\epsilon}^{L\theta} \hat{\phi}^{S\theta} / \hat{\phi}^{L\theta} \tag{40}$$

We find a idiosyncratic consumption dispersion of $\epsilon^{S+} = 3.00$ for high-income workers and $\epsilon^{S-} = 5.81$ for low-income workers. See the first panel of Table 4 for the estimated parameters. Again, high-income workers have more inelastic consumption travel patterns than low-income workers. Thus, high-income workers benefit more from reductions in travel costs as they are less able to substitute to more attractive consumption locations

[30] Since we have rich transit data, with a panel of wages across neighborhoods, we would estimate a differenced version of Equation (39), so that identification would only require stability of sorting patterns across neighborhoods. We are unable to do so because we only have wage data from 2010 and 2015, with the former being before our sample period.

in equilibrium. Idiosyncratic dispersion in consumption is larger for consumption trips than workplace trips.

6.3.3 Travel Costs

Finally, we can back out the travel cost parameter by worker type:

$$\hat{\kappa}^{\theta} = \hat{\phi}^{S\theta} / \hat{\epsilon}^{S\theta} \tag{41}$$

We estimate a travel cost parameter of $\kappa^{+} = 0.018$ for high-income workers and $\kappa^{+} = 0.014$ for low-income workers, as displayed in the first panel of Table 4. Low-income workers have slightly lower travel costs. The wedge between travel costs is consistent with the 25% subsidy low-income workers receive on their fare costs.[31]

6.4 Parameter Estimation: Dispersion of Idiosyncratic Residential Amenities and Externalities

To estimate spillovers in amenities and production, we exploit the fact that changes in market access induced by the Downtown line result in a shock to the supply of labor and resident locations across the city. To operationalize this fact, we estimate the model in 2015 and 2018 before and after the opening of the DTL.[32] By log-linearizing Equation (10) and substituting in Equation (30), we can estimate $\mu_{U\theta}$ and $\epsilon^{R\theta}$ with the following equation.

$$\Delta \ln \lambda_n^{R\theta} = \epsilon^{R\theta} \Delta \ln Q_n^{-\alpha^{H\theta}} \mathbb{W}_n^{\theta} \mathbb{S}_n^{\theta} + \epsilon^{R\theta} \mu_{U\theta} \Delta \ln(R_n^{\theta} / R_n^{-\theta}) + \Delta \ln e_{n\theta} \tag{42}$$

where Δ indicates the difference in estimated model unobservables (or data) before and after the line opening. Identification of $\epsilon^{R\theta}$ and $\mu_{U\theta}$ requires that changes in local rents and access are orthogonal to changes in latent residential amenities, controlling for residential composition.[33]

We present the estimates from Equation (42) in the second panel of Table 4. We find that dispersion of idiosyncratic residential amenities is slightly larger for high types at

[31] The ratio of travel costs between low- and high-income workers is $0.014/0.018 = 0.778$.

[32] We use housing prices from 2008, just before the precise station locations on the DTL were announced. In our descriptive analysis in Subsection 4.5, we showed that price appreciation, between control apartments and treated apartments near the planned line, diverged after 2008, but not before. Were we to use prices from 2015, much of the improved accessibility from the line would have already been "priced in."

[33] While our identification assumption is strong, our main results do not hinge on the dispersion of spillovers in amenities and production. In Table A7, we report our counterfactual estimates in a model without spillovers, finding no qualitative difference compared to our preferred specification.

$\epsilon^{R+} = 1.48$ than that of low types: $\epsilon^{R-} = 1.38$. This implies that high-income workers are better able to take advantage of improvements in access across neighborhoods by shifting where they live. We find stronger externalities for low types ($\mu_{U-} = 0.45$) than high types ($\mu_{U+} = 0.22$), consistent with the findings from Tsivanidis (2019). This implies that there are stronger spillovers in endogenous amenities for low-income workers, and hence a stronger force towards clustering in the same neighborhoods.

6.5 Estimating Residential Amenities

We define composite residential amenities $\mathbb{B}_{n\theta}$ and recover them using known parameters, observed land prices and travel shares, and estimated measures of work and consumption access.

$$\mathbb{B}_{n\theta} \triangleq B_n^{\theta}(T_n^{R\theta})^{1/\epsilon^{R\theta}} = \frac{Q_n^{\alpha^H}(\lambda_n^{R\theta})^{1/\epsilon^{R\theta}}\bar{U}^{\theta}}{\mathbb{W}_{n\theta}\mathbb{S}_{n\theta}}\Gamma\left(\frac{\epsilon^{R\theta}-1}{\epsilon^{R\theta}}\right)^{-1} \tag{43}$$

In Table 5, we show that composite residential amenities are highly correlated with external data on amenities — specifically schools, community clubs, and parks. We find that high-income residents care more about community clubs and schools overall. However, low-income residents care more about parks, as well as the presence of "neighborhood schools", public schools with generally non-legacy and non-competitive admissions.[34]

6.6 Model Inversion

The model contains unobserved endogenous variables which are necessary to conduct our counterfactuals in Section 7. We are able to recover unique values of these variables that rationalize the observed data as a model equilibrium. First of all, although we observe the workplace commuting flows between each neighborhood by worker type, we do not directly observe the share of employment in each sector, non-tradable versus tradable, conditional on residential and workplace choice by worker type. However, we are able to combine data on tradable versus non-tradable land quantities, exploiting the structure of our model, to solve for the share of employment in each sector by location and type. Second, although we observe residential land prices and commercial land quantities and rents from administrative data sources, we do not observe residential land quantities. However, we can use the model's structure to recover residential land quantities by dividing total housing expenditures for all the residents in neighborhood by residential

[34] Wealthier families typically can send their children to elite and/or private schools in which they have legacy status.

land prices, as in Equation (23). Thus it suffices to observe data on residential wages and population by type. The following proposition formalizes the intuition behind our model inversion process.

Proposition 1. ***Sector Employment by Type and Neighborhood*** *(i) Given data on residence and work travel by type and commercial land quantities by sector* $\{\lambda^{L\theta}_{ni|n}, \lambda^{R\theta}_{n}, R^{\theta}, H_{nG}\}$ *in addition to model parameters, there exists a unique vector of neighborhood-sector employment shares conditional residence by type,* $\{\lambda^{L\theta}_{niG|n}\}$*, that rationalizes the observed data as an equilibrium of the model.*

Residential Quantities *(ii) Given data on residence and wages by type, and residential land prices* $\{\lambda^{R\theta}_{n}, R^{\theta}, \mathbb{W}^{\theta}_{n}, Q_n\}$ *in addition to model parameters, there exists a unique vector of residential land quantities,* $\{H^{R}_{n}\}$*, that rationalizes the observed data as an equilibrium of the model.*

6.7 Calibrated Parameters

Calibrated values are reported in Table A6 in the Online Appendix. The consumption shares, α, for low- and high-income groups (below and above the 25th percentile) are calibrated to match line-itemized expenditures data from the 2018 Singapore Household Expenditure Survey. These expenditures data are available by income bracket. The overall labor share by sector, β^{G}, is matched to estimates from the 2013 Economic Survey of Singapore. Labor shares by type, $\beta^{G\theta}$, are calibrated to the average share of the wage bill paid to low- and high-income workers in Singapore by sector using the 2018 Singapore Labor Force Survey. We assume a share of land in construction costs of $\mu = 0.25$ following Epple, Gordon, and Sieg (2010), Combes, Duranton, and Gobillon (2019) and Ahlfeldt et al. (2015). We consider production agglomeration parameters consistent with Rosenthal and Strange (2004), Melo et al. (2009), and Ahlfeldt et al. (2015). In our preferred specification, we set the productivity elasticity μ_A to be 0.03.

7 Counterfactuals

In this section, we describe the counterfactual scenarios we study. In our main analysis, we compute the impact of the Downtown Line on inequality and welfare.[35] To assess the robustness of our conclusions to alternative transit placements, we will also consider the effect of removing two different sets of metro lines.

[35]For data availability reasons, we focus on Phase 2 and 3 of the line. These two phases comprise 80% of the line and connect the outer neighborhoods of the city to downtown (as opposed to Phase 1, a short "test" segment downtown).

This section proceeds as follows. First, we outline our exact-hat approach. Second, we test our counterfactual predictions on ex-post data. Finally, we present our main results.

7.1 Exact-hat

We undertake counterfactuals with an exact-hat approach (Dekle, Eaton, and Kortum 2008). Rather than estimating our model in terms of levels, we specify the model in terms of changes from the current equilibrium, thereby finessing having to assemble proxies for various unobservables in our model.[36] We let $\hat{x} = x'/x$ denote the relative change in the equilibrium endogenous variable x, where x' is the value of x in the new equilibrium. We consider an exogenous change in travel times $\Delta\tau_{ij} = \tau'_{ij} - \tau_{ij}$. Given model parameters $\{\alpha, \beta, \kappa^{\theta}, \epsilon^{R\theta}, \epsilon^{L\theta}, \epsilon^{S\theta}, \mu_H, \mu_A, \mu_{U\theta}\}$, assumed bilateral changes in travel times, $\{\exp(\Delta\tau_{ij})\}$, and endogenous variables in the initial equilibrium $\{\lambda^{S\theta}_{nj|n}, \lambda^{L\theta}_{ni|n}, \lambda^{R\theta}_{n}, H_{nG}, H^{R}_{n}, \mathbb{W}\}$, we solve the system of "hat equations"[37] starting with an initial guess in each endogenous variable such that $\hat{x} = 1$, updating our guess till the algorithm has converged to an equilibrium. With the counterfactual changes in endogenous variables $\{\hat{B}^{\theta}_{n}, \hat{\mathbb{S}}^{\theta}_{n}, \hat{\mathbb{W}}^{\theta}_{n}, \hat{Q}_{n}\}$, the change in expected utility by type is:

$$\hat{\tilde{U}}^{\theta} = \left(\sum_{n} \lambda^{R\theta}_{n} \left(\hat{B}^{\theta}_{n} \hat{\mathbb{S}}^{\theta}_{n} \hat{\mathbb{W}}^{\theta}_{n} \hat{Q}_{n}^{-\alpha^{H}} \right)^{\epsilon^{R\theta}} \right)^{1/\epsilon^{R\theta}} \tag{44}$$

We estimate the initial equilibrium using data from 2015, then use changes in travel times as observed in the fare card data from before the opening of the DTL in 2015 to after its opening in 2018. Figure 10 presents the distribution of the changes in travel time.

In Online Appendix E.1, we test our model's predictions on data from 2018. We find that predicted changes in travel patterns and residential shares are highly correlated with observed post-DTL changes. Furthermore, predicted changes in land use for non-tradable good production are correlated with post-DTL food establishment entry between 2015 to 2018. Thus, the model has predictive power over changes in travel patterns, residence choices and commercial activity in response to the DTL.

[36] These unobservables include the prices of non-tradable goods, neighborhood-sector wages per efficiency unit by type, mean idiosyncratic draws, and so on.

[37] Equations are available in Online Appendix D.

7.2 Welfare and Inequality

7.2.1 Main Results

Column 1 of Table 6 presents the effect of the Downtown line on welfare, broken down by access to employment and consumption by high- and low-income types respectively. We find that the DTL improves welfare, $\bar{U}^{\theta}$, for high income workers, but not for low-income workers. Welfare increases by 3.15% for high-income workers, while welfare for low-income workers remains stagnant at a 0.04% increase. Although consumption access increases for both groups, non-tradable production and low-income jobs move to less attractive locations. Access to consumption, $\mathbb{S}$, improves for both workers, with high income workers benefiting by 1.96%, higher than low-income workers at 1.55%. However, while access to employment (expected wages net of commuting costs), $\mathbb{W}$, increases for high-income workers by 0.77%, low-income workers experience a 1.65% decline. Overall, inequality increases by 3.11%.[38]

There are two main mechanisms driving our results. First, the DTL disproportionately improves access to residents of many high-income areas. Figure 1 shows that the Downtown Line directly serves the many high income areas in a cluster just west of the city center. This partially explains the larger improvements in both access to consumption and employment experienced by high-income workers relative to low-income workers. Second, in response to the DTL, a greater number of consumers travel to consume non-tradables near the center of the city. These consumers benefit from improved access to downtown via the new line. Figure 11 plots the model predictions for changes in consumption trips across neighborhoods. We see a large increase in consumption trips made to the center of the city. As a result, non-tradable jobs, disproportionately low-income, move to the center of the city; meanwhile, low-income workers live far away from the center, as seen in Figure 1. Indeed, average commute times increased 1.5% for low-income workers, while high income workers saw a 1% reduction. Consistent with this finding, we estimate an increase in segregation as measured by the dissimilarity index.[39]

7.2.2 Results with Only Work Travel

We see much larger inequality effects by including travel to consume non-tradables. To see how our results change when consumption travel is not accounted for, we set $\alpha^{N,\theta} = 0$, $\alpha^{T,\theta} = 1 - \alpha^{H,\theta}$ and $\lambda^{S\theta}_{ni|n} = 0$ for both worker groups. Column 2 of Table 6 presents the

[38] Our results are robust to the absence of spillovers, as well as the "covariates-based approach" of Dingel and Tintelnot (2020). See Online Appendix E for details.

[39] Our measure of dissimilarity is $\frac{1}{2}\sum_n |\frac{R^+}{R^-} - \frac{R^+\lambda^{R+}_n}{R^-\lambda^{R-}_n}|$.

results from the re-estimated model. We find that the DTL improves welfare, $\bar{U}^{\theta}$, for both worker groups. Welfare increases by 1.44% for high-income workers, while welfare increases by 0.84% for low-income workers. Expected income net of commuting, or access to employment, increases by 1.43% for high-income workers and by 0.43% for low-income workers. We do not capture improvements in consumption access. Overall inequality only increases by 0.60%, a five-fold underestimation relative to our baseline results.

When estimated without consumption trips, the model understates the welfare effects of the DTL because it ignores the fact that workers value access to consumption. Furthermore, the model also severely underestimates the inequality effects of the line in two key ways. First, the model fails to capture the disproportionate gains in access to consumption for high-income workers. Second, this version of the model misses the re-organization of non-tradable production across the city in response to changes in consumption travel, worsening access to employment for low-income workers.

7.2.3 Impacts of Other Lines in Singapore

In our main results, we find that the poor experience almost no welfare gains from the DTL, driven by reductions in employment access. Do these results generalize to other train lines? We consider two auxiliary counterfactuals. First, we remove the North South Line (NSL), one of the two original train lines in the Singapore metro system. The NSL is a trunk line primarily serving the northern neighborhoods of Woodlands.[40] We feature this line because the suburban neighborhoods it serves are much more socioeconomically diverse than those of the DTL. Column 1 of Table A9 displays the associated results. When the NSL is removed, we predict that welfare will fall for both low and high types, by 4.41% and 2.08% respectively. Both types experience reductions in consumption and employment access, and inequality rises by 2.33%. Our takeaway is that line placement matters for understanding the impacts of transit expansion on welfare and equity.[41]

Second, in a stylized exercise, we consider what happens if the entire rail system never existed, maintaining the current level of bus service. Column 2 of Table A9 displays the associated results. We predict that welfare falls by about 15% for both types, driven

[40] The NSL begins in the west-southwestern suburbs of Jurong, travels due north through Woodlands and then veers due south into downtown.

[41] Can the alignment of the DTL be improved? We discuss two alternative line placements, each with their own flaws. The DTL takes the most direct route from the Bukit Panjang suburbs to the downtown area, passing by the wealthy neighborhoods of Bukit Timah. A "northern" candidate would run underneath the Central Catchment Area and meet the Circle Line at Caldecott Station, incurring high construction costs from tunnel construction under a large water body. A "southern" candidate connects to the East West Line at Dover and/or the Circle Line at one-north before continuing downtown, duplicating the (wealthy) neighborhoods served by the latter two lines.

by reductions in consumption and employment access of about 6% and 7%. We view our results as suggestive evidence of the robustness of Singapore's (bus) transit system, though the additional funds saved from not building rail could be used to bolster the bus and road network. The welfare improvements from an optimal urban transit network in Singapore is an open question (see e.g. Fajgelbaum and Schaal 2020).

7.3 Decomposition

We conduct a model decomposition exercise, detailed in Online Appendix E.4, that aims to isolate how various differences across worker groups impact inequality. First, low-income workers spend less on non-tradables than the high-income. Thus, they experience smaller gains in welfare from better being able to access consumption opportunities. Second, since low-income workers have slightly lower travel costs than high-income workers, they gain less from shorter travel times. Third, higher travel elasticities imply that low-income workers can, in equilibrium, better substitute to attractive work and consumption locations. Therefore, they benefit less from lower travel times. Lastly, since low-income workers have relatively more-dispersed residential preferences, they are less able to move to take advantage of improved access elsewhere.

8 Conclusion

This paper demonstrates that consumption trips have important implications for the inequality effects of public transit expansions. We develop an urban spatial model with heterogeneous worker groups, low- and high-income, which incorporates both commuting and travel to consume non-tradable goods and services. We use the model to study the impact of the Downtown Line (DTL) in Singapore. We find that the DTL improves welfare for high income workers by 3%. Abstracting away from travel to consume non-tradables results in a five-fold underestimation of the inequality effects of the DTL. That is, absent consumption travel, we find that both worker groups benefit, but a slightly larger share of the gains goes to high-income workers. This underestimation follows because low-income non-tradable sector jobs reorganize spatially. Aggregate welfare gains are also underestimated by about 40%, ignoring gains in access to consumption opportunities for both groups.

Our results sound a note of caution to a transit planner: line placement matters. The DTL links a mixed-income neighborhood to the city center through wealthy neighborhoods. When similar lines serving more diverse neighborhoods are removed, inequal-

ity rises. Policymakers should consider the welfare and distributional implications of changes in consumption patterns, in addition to commuting, as they plan their mass transit systems. We invite future research in other urban settings, as well as in the optimal design of transit networks and (co-)location of housing more generally.

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Figures and Tables

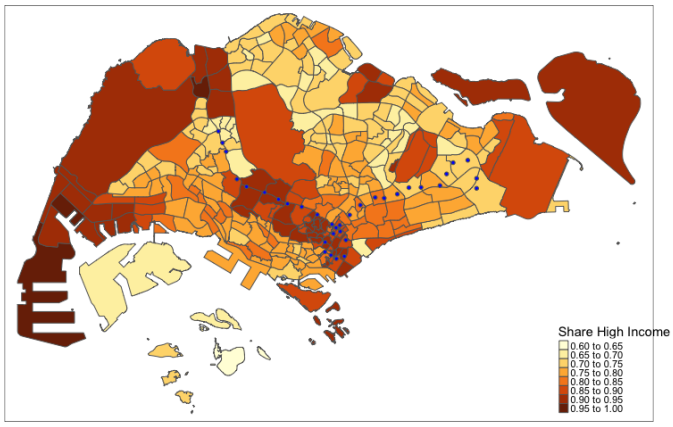


Figure 1: High-Income Residential Population Share and the Downtown Line

Notes: This figure plots the share of high-income residents in each neighborhood, or subzone, according to farecard data in 2015. We define high-income residents to be those earning above the 25th percentile. The Downtown line stations are plotted in blue.

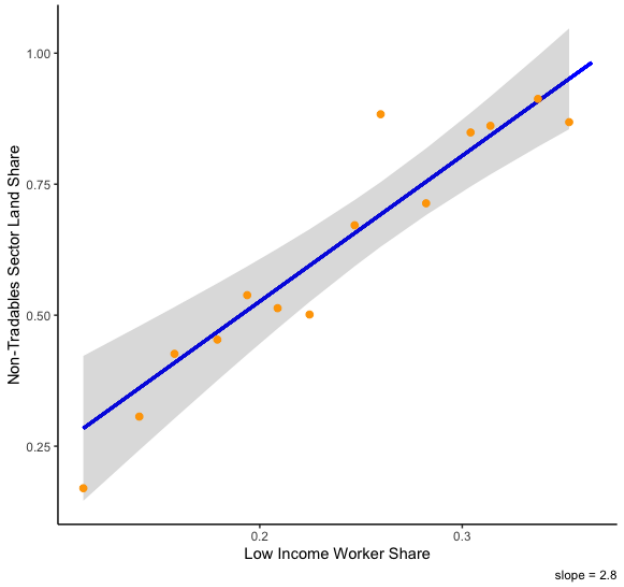
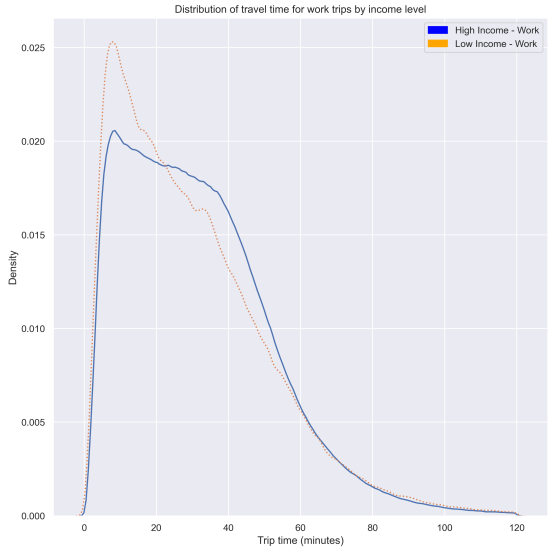
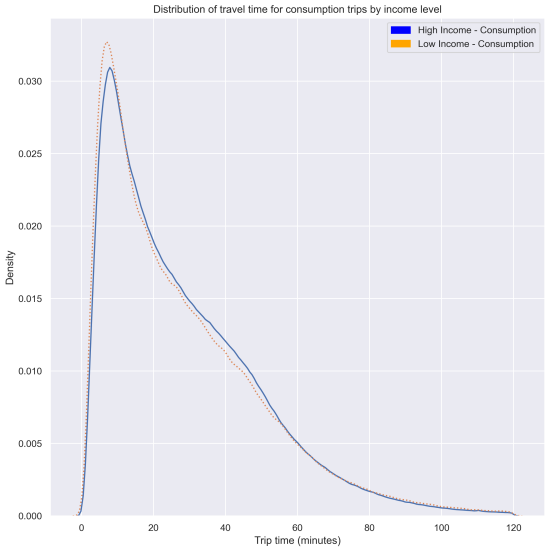


Figure 2: Relationship Between Low-Income Employment and Commercial Land Share for Non-Tradables

Notes: This figure displays a binned scatter plot of the share of low-income (below the 25th percentile) employment against the share of commercial land used by non-tradable sectors by neighborhood in 2015. Each point shows the mean of the variable with low-income employment share in the given bin.



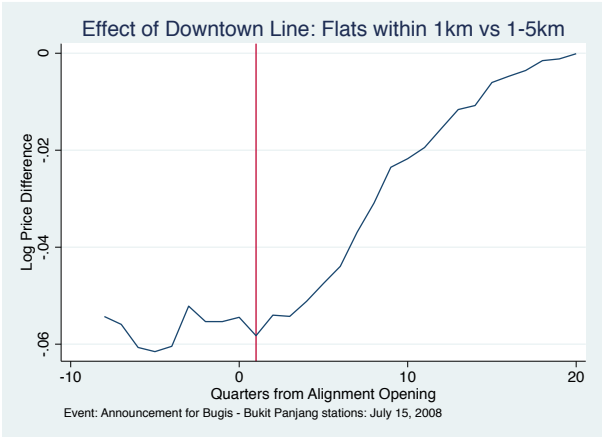
(a) Work



(b) Consumption

Figure 3: Distribution of Travel Time

Notes: This figure presents the distribution of travel times for work in panel (a) and for consumption in panel (b) by high- and low-income workers respectively (above and below the 25th percentile). Travel time is computed at the trip level from farecard data pooled across all years.



(a) Spatial Difference-in-Differences

(b) Prices vs. Distance to DTL Over Time

Figure 4: Relationship between Prices and Distance to the Downtown Line Over Time

Notes: The left panel plots the log difference in mean residential prices between government apartments within 1km and between 1-5 km of a Phase 2 DTL station in each quarter, relative to the announcement date of the alignment of the line on 15 July 2008 (indicated with a red vertical line). The right panel plots the relationship, over time, between log apartment prices and the distance (in km) from the DTL. We estimate Equation (46) and plot the coefficients γ from that regression. Apartment transactions are grouped by quarter, from 2007 to 2011, relative to the announcement date of the Phase 2 alignment of the line on 15 July 2008. We restrict our sample to addresses within 5km of a Phase 2 DTL station. Our data are a geocoded balanced panel of resale transactions for Housing and Development Board flats in Singapore.

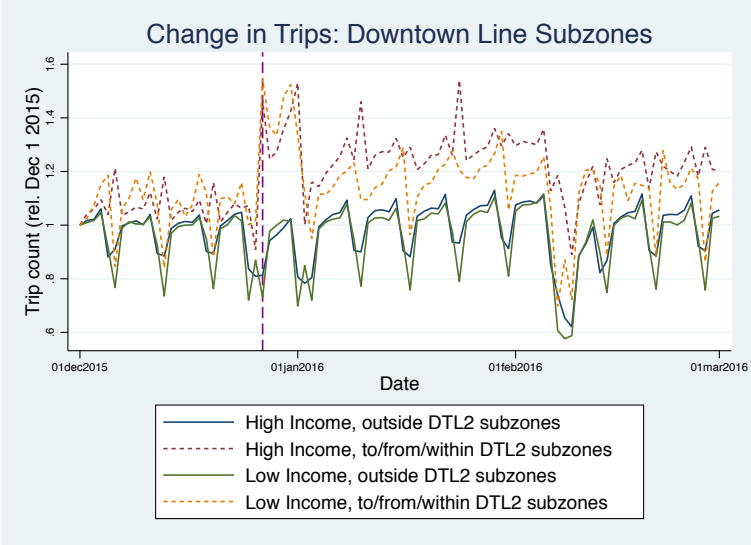
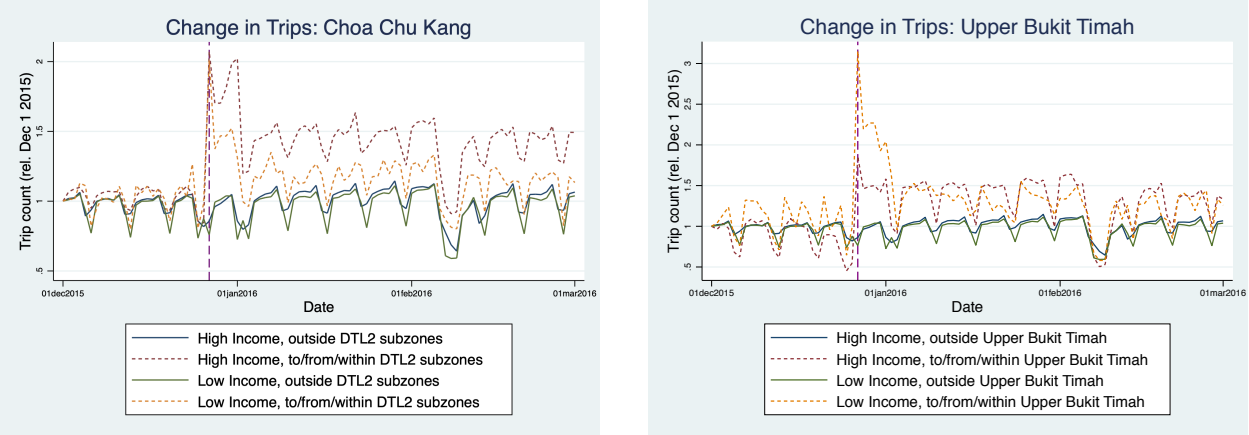


Figure 5: Impact of the Downtown Line on Travel by High- and Low-Income Workers

Notes: This figure plots the daily volume of trips to and from subzones with and without a DTL station over time, separated by high- and low-income workers (above and below the 25th percentile respectively). Fare card data are from December 2015 to February 2016. The number of trips is normalized with respect to that of Dec 1 2015.



(a) Choa Chu Kang

(b) Upper Bukit Timah

Figure 6: Impact of the Downtown Line on Travel by High- and Low-Income Workers: By Station

Notes: This figure plots daily volume of trips to and from Choa Chu Kang in Panel (a) and Upper Bukit Timah in panel (b), compared to all other stations over time, by high- and low-income workers (above and below the 25th percentile). Fare card data are from December 2015 to February 2016. The number of trips is normalized with respect to that of Dec 1 2015.

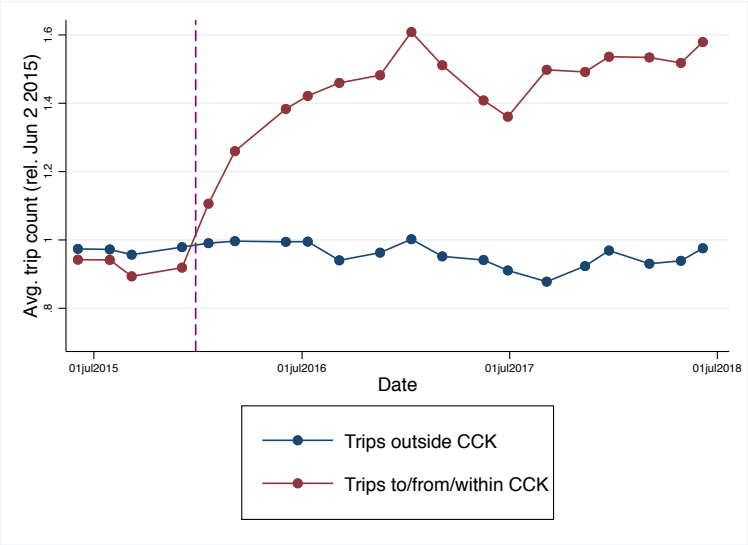
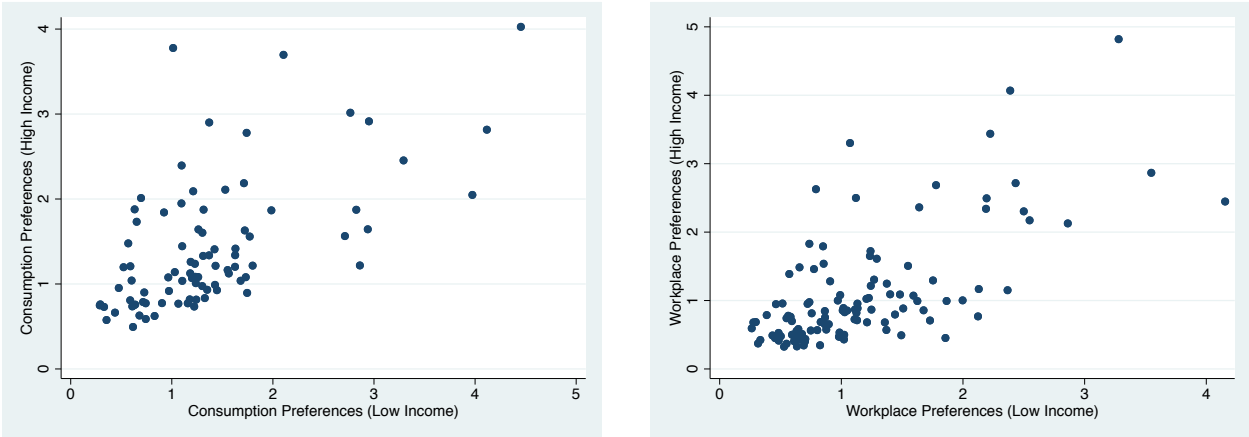


Figure 7: Impact of the Downtown Line on Long-run Travel: Choa Chu Kang (CCK) Stations

Notes: This figure plots the quarterly volume of trips to and from Choa Chu Kang, compared to trips originating and ending at all other stations, by high- and low-income workers (above and below the 25th percentile). The plot is made with fare card data between July 2015 and July 2018. We plot the number of trips relative to the number in June 2015.



(a) Consumption (b) Work

Figure 8: Work and Consumption Location Attractiveness by High- and Low-Income Workers

Notes: This figure plots consumption location attractiveness in Panel (a) and workplace attractiveness in Panel (b). Location attractiveness measures arise from estimation in Section 6, and differ between high- and low-income workers and at the subzone level.

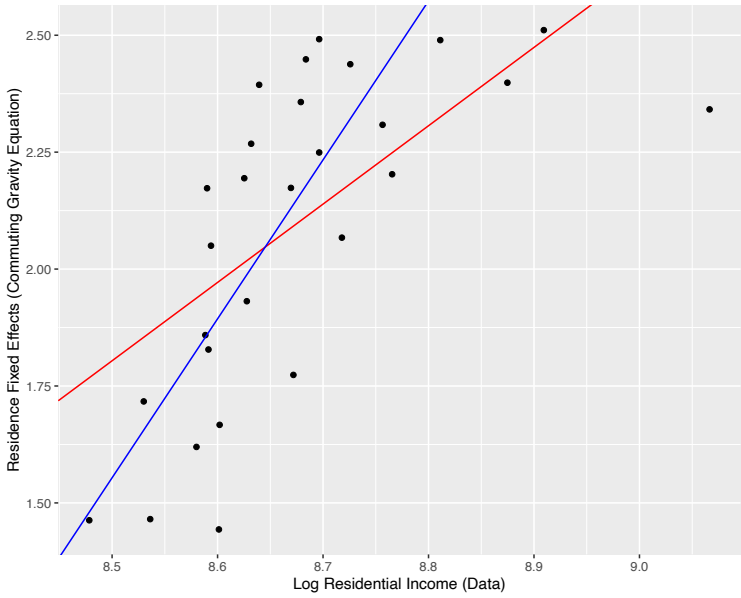


Figure 9: Expected Income: Data vs Model

Notes: This figure displays a binned scatter plot of log model-estimated expected income by residence against log observed data on income by residence in 2015, from the Singapore General Household Survey in that year. Each point shows the mean of the variable with observed data on income in the given bin. The red line is the linear fit across all neighborhoods, and the blue line is the linear fit dropping the 4 neighborhoods with the highest income.

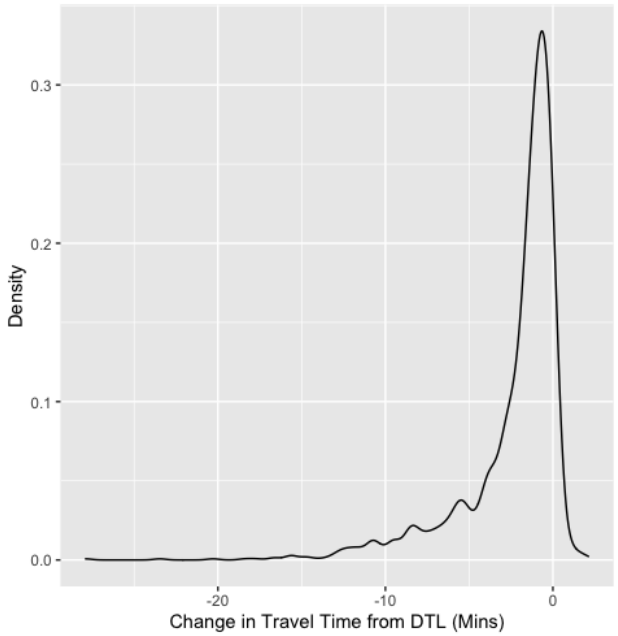


Figure 10: Changes in Bilateral Transit Time: Downtown Line

Notes: This figure displays the distribution of the changes in bilateral travel times in minutes across all neighborhoods. The span of the fare card data are from 2015 to 2018.

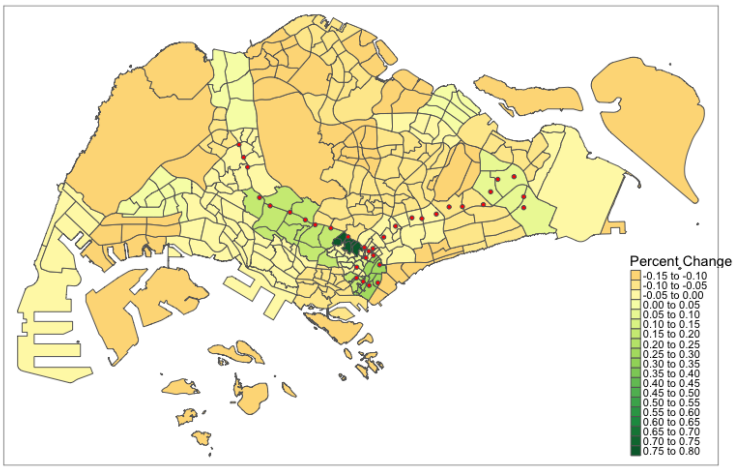


Figure 11: Model Prediction: Change in Consumption Trips

Notes: This figure plots the predicted changes in total consumption trips in the counterfactual estimation in Section 7.

Table 1: Gravity Estimation

	Travel Share Conditional on Residence			
	Commuting (High Income)	Commuting (Low Income)	Consumption (High Income)	Consumption (Low Income)
	(1)	(2)	(3)	(4)
Travel Time (Minutes)	-0.051^{***}	-0.070^{***}	-0.053^{***}	-0.094^{***}
	(0.001)	(0.002)	(0.001)	(0.004)
Dest. FE	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes
IV (geog. dist)	Yes	Yes	Yes	Yes
R^2	0.854	0.660	0.792	0.576
Adjusted R^2	0.846	0.643	0.781	0.554

*p<0.1; **p<0.05; ***p<0.01

Notes: This table displays results from estimating Equation (31) in Columns 1 and 2, and Equation (35) in in Columns 3 and 4, for high- and low-income workers respectively. Destination and origin fixed effects are included, as well as an instrument, for mean travel time, of straight-line distance between neighborhood centroids. Each observation is a bilateral share of travel to a destination, conditional on residence, from 2015 fare card data.

Table 2: Consumption Location Attractiveness: Data vs Model

	Log Consumption Attractiveness:									
	High Income	Low Income	High Income	Low Income	High Income	Low Income	High Income	Low Income	High Income	Low Income
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Food Establishments	0.386^{***}	0.493^{***}								
	(0.036)	(0.073)								
Hawker Stalls / Food Establishments			0.051	0.188^{**}						
			(0.043)	(0.076)						
Food Court / Food Establishments					0.080^{*}	0.295^{***}				
					(0.043)	(0.075)				
Supermarkets							0.415^{***}	0.586^{***}		
							(0.035)	(0.071)		
Clinics									0.424^{***}	0.733^{***}
									(0.035)	(0.067)
R^2	0.277	0.131	0.005	0.021	0.012	0.051	0.319	0.184	0.333	0.288
Adjusted R^2	0.274	0.128	0.001	0.017	0.008	0.048	0.317	0.181	0.331	0.286
Residual Std. Error	0.625	1.275	0.732	1.294	0.729	1.274	0.607	1.235	0.600	1.154

*p<0.1; **p<0.05; ***p<0.01

Notes: This table displays regression estimates of model-consistent measures of consumption attractiveness on various auxiliary data on retail amenities. We consider estimated high-income consumption attractiveness in odd columns and low-income consumption attractiveness in even columns, as measured in Section 6. Each observation is a neighborhood (subzone). Columns 1 and 2 consider the normalized total number of food establishments from Singapore Food Establishment data. Columns 3 and 4 only consider food establishments that are hawker stalls. Columns 5 and 6 only consider food establishments are food courts. Columns 7 and 8 consider the normalized total number of supermarkets from National Environment Agency data. Finally, columns 9 and 10 consider the normalized total number of clinics from Ministry of Health data.

Table 3: Workplace Attractiveness, Employment Density and Idiosyncratic Productivity by Worker Type

	Log Workplace Attractiveness	
	High Income	Low Income
	(1)	(2)
Employment Density (Normalized)	0.379***	0.228***
	(0.053)	(0.056)
R^2	0.143	0.052
Adjusted R^2	0.141	0.049
Residual Std. Error	0.927	0.975

*p<0.1; **p<0.05; ***p<0.01

(a) Workplace Attractiveness and Employment Density

	Parameter			
	ϵ^{L+}		ϵ^{L-}	
	(1)	(2)	(3)	(4)
Log Income (High Type)	2.075***	2.912***		
	(0.482)	(0.641)		
Log Income (Low Type)			3.637**	5.023***
			(1.410)	(1.380)
Exclude Outliers	No	Yes	No	Yes
R^2	0.328	0.378	0.149	0.280
Adjusted R^2	0.310	0.359	0.127	0.259
Residual Std. Error	0.384	0.380	0.527	0.489

*p<0.1; **p<0.05; ***p<0.01

(b) Productivity Dispersion

Notes: The left subtable displays regression estimates of model-consistent measures of workplace attractiveness on employment density. We consider estimates of high-income workplace attractiveness in Column 1 and estimates of low-income workplace attractiveness in Column 2. Workplace attractiveness is measured in Section 6. Each observation is a neighborhood (subzone). Employment density is normalized and is computed by dividing total employment (based on farecard data) and total land area of the subzone (in square meters). The right subtable displays estimation results for Equation (39) in Section 6. The dependent variable is $\epsilon^{L\theta}$ multiplied by expected income by residence, as estimated in the model, and the explanatory variable is observed income by residence from the 2015 Singapore General Household Survey. We consider high-income workers (above the 25th percentile) in Columns 1 and 2, and consider low-income workers (below the 25th percentile) in Columns 3 and 4. In Columns 2 and 4, we drop the four neighborhoods with the highest income.

Table 4: Dispersion of Idiosyncratic Consumption Amenities and Travel Cost Parameters (left) and Residential Amenities and Externalities (right) by High- and Low-Income

Parameter	**Estimate**
ϵ^{S+}	3.00
ϵ^{S-}	5.81
κ^{+}	0.018
κ^{-}	0.014

(a) Consumption Amenities and Travel Cost Parameters

	Dependent variable:	
	$\Delta \ln \lambda_n^{R+}$	$\Delta \ln \lambda_n^{R-}$
	(1)	(2)
$\Delta \ln Q_n^{-\alpha^{H+}} \mathbf{W}_n^{+} \mathbf{S}_n^{+}$	1.481***	
	(0.407)	
$\ln(R_n^{+}/R_n^{-})$	0.332***	
	(0.085)	
$\Delta \ln Q_n^{-\alpha^{H-}} \mathbf{W}_n^{-} \mathbf{S}_n^{-}$		1.380***
		(0.494)
$\ln(R_n^{-}/R_n^{+})$		0.614***
		(0.085)
R^2	0.298	0.373
Adjusted R^2	0.283	0.359
Residual Std. Error	0.237	0.245

*p<0.1; **p<0.05; ***p<0.01

(b) Residential Amenities and Externalities

Notes: The left subtable displays estimates of parameters governing the dispersion of idiosyncratic consumption amenities in rows 1 and 2 from Equation (40), as well as travel costs in rows 3 and 4 from Equation (41). Rows 1 and 3 are estimates for high-income workers (above the 25th percentile), while rows 2 and 4 are their counterparts for low-income workers (below the 25th percentile). The right subtable displays estimates of Equation (42). Column 1 considers high-income workers (above the 25th percentile), and Column 2 considers low-income workers (below the 25th percentile). The dependent variable is the change in residential share by worker type between 2015 and 2018. The explanatory variables are changes in model estimates and observed data between 2015 and 2018. Each observation is a neighborhood (subzone).

Table 5: Residential Amenities: Data vs Model

	Log Residential Amenities:							
	High Income	Low Income	High Income	Low Income	High Income	Low Income	High Income	Low Income
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Parks	0.174***	0.220***						
	(0.054)	(0.058)						
Community Clubs			0.408***	0.400***				
			(0.059)	(0.073)				
All Schools					0.446***	0.399***		
					(0.062)	(0.081)		
Neighborhood Schools							0.196**	0.216**
							(0.077)	(0.088)
High-Low Pop. Ratio Control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.131	0.531	0.300	0.574	0.313	0.559	0.180	0.331
Adjusted R^2	0.118	0.525	0.290	0.568	0.303	0.552	0.128	0.289
Residual Std. Error	0.870	0.945	0.781	0.901	0.773	0.917	0.456	0.518

*p<0.1; **p<0.05; ***p<0.01

Notes: This table displays regression coefficients validating model amenities with auxiliary data. Model-consistent measures of residential amenities are regressed on, separately, the number of parks, community clubs, all schools and "neighborhood" schools in each neighborhood. We consider estimated high-income composite residential amenities in odd columns and low-income composite residential amenities in even columns. Model amenities are measured in Section 6. Each observation is a neighborhood (subzone). Columns 1 and 2 consider the normalized total number of parks from Singapore Land Authority data. Columns 3 and 4 consider the normalized total number of community clubs from Singapore Land Authority data. Columns 5 and 6 consider the normalized total number of schools from Ministry of Education data. Columns 7 and 8 consider the normalized total number of "neighborhood schools" from and defined by Ministry of Education data. "Neighborhood schools" are public schools which are non-legacy and do not have gifted education programs.

Table 6: Impact of the Downtown Line: Main Estimates

	(1) Full Model		**(2) Only Commuting**	
% Change in	Low Type	High Type	Low Type	High Type
Welfare, $\bar{U}$	0.04	3.15	0.84	1.44
Access to Consumption, $\mathbb{S}$	1.55	1.96	0	0
Access to Employment, $\mathbb{W}$	-1.65	0.77	0.43	1.43
Inequality	3.11		0.60	
Segregation	2.70		1.79	

Notes: This table displays results from the counterfactual estimation in Section 7. Column 1 displays results from the full model. Column 2 shows results abstracting away from consumption trips. Welfare, access to consumption and access to employment are defined by the model in Section 5. Inequality is the difference in percentage changes in welfare across high- and low-income workers. Segregation is measured by the dissimilarity index.